

TABLE 25-1 Quality Assurance Tests for MRI Systems

Frequency	Tests
Daily	Resonance frequency Signal-to-noise ratio
Monthly	Image uniformity Linearity, hard copy
Yearly	Magnetic field homogeneity RF pulse shape and amplitude Gradient pulse shape and amplitude

while the signal from water is shifted in the opposite direction. If this shift exceeds the size of a pixel in the frequency-encoding dimension, then the signals from these substances are erroneously placed in the image matrix.

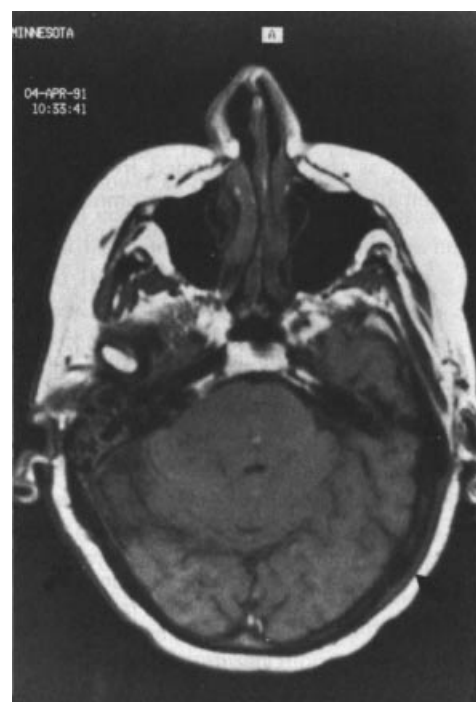
■ QUALITY ASSURANCE

Quality assurance in MRI reflects the principles of image formation and display that determine the appearance of any medical image. Tests such as resolution, contrast, linearity, and sensitivity are a necessary part of evaluating equipment performance. The actual testing methods and the standards for performance are, of course, unique to MRI. There are two approaches to quality assurance measurements. Direct measurement of image quality through evaluation of end points such as imaging system uniformity is essential. Some imaging centers go further: they measure intermediate steps in the imaging process such as tip angle or gradient pulse shape to identify problems before they produce noticeable effects on image quality. A number of guides now exist for quality assurance in MRI.^{8–10} These guides cover the range and recommended frequency of quality assurance measurements of these measurements. Table 25-1 summarizes some of these recommendations.

■ BIOEFFECTS

The US Food and Drug Administration has published guidelines for safe operating characteristics of MRI systems.¹¹ Manufacturers must follow these guidelines in order to obtain approval to market MRI devices. Exemptions to the guidelines require submission to the FDA of scientific evidence verifying the safety of patients if the guidance levels are exceeded. Under FDA guidance for clinical MRI, the maximum static magnetic field strength is 4 tesla and the maximum acoustic noise level (caused by thermal and mechanical stress in the magnet due to gradient switching) is 140 dB. The specific energy absorption rate and the time rate of change of the magnetic fields are addressed in three categories as “normal mode,” “first level controlled,” and “second level controlled.” Normal mode refers to routine operation for patient studies. In first level controlled mode, a clear indication that the unit will operate in this mode must be visible to the operator and a positive action must be taken by the operator to initiate the scan. In second level controlled mode, security measures such as a key lock or a software password must be used. These parameters are described in Table 25-2.

At the present time, there is no conclusive evidence for adverse biologic effects in normal patients scanned in MRI systems using parameters that have been approved by the FDA. There are areas of concern, however, that are being monitored in a number of studies. Some of these studies are described in the following section. Areas of concern are categorized according to the main physical features of MRI, namely, the static magnetic field, time-varying magnetic field, and RF energy.

**MARGIN FIGURE 25-9**

A chemical shift artifact produced by the difference in resonance frequency between lipids and water. The frequency difference affects spatial encoding and causes the lipid signal to be placed in the image a short distance away from the water compartment of the same tissue.

TABLE 25-2 FDA Safety Guidelines for Specific Absorption Rate (SAR) and Time Rate of Change of Magnetic Field (dB/dt) in Commercially Manufactured MRI Systems

Specific Absorption Rate ^a (SAR)	Watt/kg		
	Normal Mode	First Level Controlled	Second Level Controlled
Whole body, averaged over 15 minutes	1.5	4	10
Head, averaged over 10 minutes	≤3	None	≥3
Local tissue, averaged over any gram of tissue over 5 minutes			
Head or torso	≤8	None	≥8
Extremities	≤12	None	≥12
Time rate of change of magnetic field (dB/dt)	20 T/sec	Mild peripheral nerve stimulation ^b	Painful peripheral nerve stimulation ^b

^aAlternatively, evidence may be submitted that RF energy levels will not cause more than 1°C rise in core body temperature, or localized temperatures in excess of 38°C in the head, 39°C in the trunk, and 40°C in the extremities.

^bDetermined by experimentation with at least 20 volunteer subjects.

A visual phenomenon may be produced if patients move their heads rapidly in a magnetic field above about 10 millitesla. This phenomenon, a visual sensation of flashing lights known as magnetophosphenes, is not thought to affect vision permanently and ceases immediately following magnetic field exposure. It was first discovered by d'Arsonval in 1896 and has been described in MRI systems since the mid-1980s.¹²

Static Magnetic Field

Some creatures such as mollusks and bees are known to be sensitive to magnetic fields. Such effects are caused by the presence of magnetic receptors in a sensory apparatus that contains small amounts of naturally occurring ferromagnetic materials. The migration patterns of birds, for example, are known to be affected by variations in the strength of the earth's magnetic field that are detected by receptors embedded in the cranium. Analogous structures have not been identified in humans. To date, the only significant biologic effect of static magnetic fields on higher organisms is the induction of electrical potentials on the order of millivolts due to motion of blood through a magnetic field. This is not known to have any adverse consequences.

Epidemiologic studies have been conducted in humans exposed to a wide range of static magnetic field strengths. Most of these studies are related more to fields experienced by workers in the control area or by technologists in the magnet room during patient setup than to the magnetic fields experienced by patients. Studies of workers who use electrolytic cells for chemical separation and experience fields typically below 15 millitesla, and of workers at national research laboratories who are exposed to field strengths of 0.5 millitesla for long periods of time or up to 2 tesla for short periods, have yielded no significant increases in disease rates.^{13,14} Workers exposed to magnetic fields (less than 56 millitesla) in the aluminum industry have been found to be at greater risk for leukemia.^{15,16} However, a wide spectrum of environmental contaminants is also present for these workers. At the present time, no causal relationships have been proposed.¹⁷ The FDA has published clinical exposure guidelines for limiting human exposure to a maximum value of 4 tesla (Table 25-2). This upper bound represents the currently accepted limit for safe general usage, but is not thought to represent an absolute threshold of any kind.

Time-Varying Magnetic Fields

Electrical fields and currents may be induced in a conductor such as the human body when magnetic fields are changed (increased or decreased in magnitude) in the vicinity of the conductor. The principle behind this phenomenon, called Faraday's law, is the same as that responsible for induction of a current (the MR signal) in a receiver coil by precessing nuclei in the sample.

Small electrical currents are induced by time-varying gradient magnetic fields in MRI units. Gradient changes on the order of a few tesla per second are typical maximum values encountered during 2DFT clinical imaging techniques. These rates of change of the magnetic field induce currents on the order of a few tens of milliamperes

per square meter. To put this figure into perspective, one should note that currents of up to 10 mA/m^2 are routinely present in brain and heart tissue.

Some fast imaging schemes may lead to larger induced current densities. The number of gradient-switching operations per unit time is not the only determinant in this regard. The instantaneous rate of change of the magnetic field during any single gradient-switching operation determines the maximum induced current, while the total number of such operations determines the duration.

Induced current densities of 10 to 100 mA/m^2 have been shown to produce permanent changes in cell function if the exposure is chronic. Examples of these changes include cell growth rate, respiration and metabolism, immune response, and gene expression. A bibliography and summary of these studies is given in work by Tenforde and Budinger.¹⁷

Whether deleterious or beneficial, the occurrence of biologic effects at current densities exceeding 10 mA/m^2 suggests the need for further research. Various mechanisms have been proposed for the cell response to induced electrical fields and their associated currents at levels encountered by MRI patients. These models suggest that membrane effects are plausible,¹⁹ even at electrical field levels that are below the random background electronic “noise” at the cell surface.^{20,21}

Current densities above 1000 mA/m^2 (1 A/m^2) produce irreversible deleterious effects such as cardiac fibrillation, stimulation of seizures, and respiratory tetanus. These current densities are orders of magnitude above those induced by clinical MRI systems used at the present time. It is difficult to produce such current densities in biologic specimens without the implantation of electrodes directly into the tissue.

Rates of change of gradient field strength used for imaging are the same order of magnitude as the changing magnetic field component of extremely low frequency (ELF) electromagnetic fields such as those associated with municipal transmission and distribution electrical lines. Such effects have received attention as a result of epidemiologic studies that suggest weak links with deleterious health effects. While laboratory studies involving human volunteers have failed to demonstrate any deleterious effects, more than a dozen epidemiologic studies during the decade 1981 to 1990 suggested an increased probability of cancer for individuals living near electrical power lines.^{22–27} Residential power lines operate at frequencies of 60 Hz, which qualifies as ELF, and induce currents in humans that range from 10^{-3} to 1 mA/m^2 . Although some studies contain serious design flaws such as unmatched controls, failure to establish a dose–response relationship, failure to control for the presence of known carcinogens, and so on, some suggestive data remain. The electrical field strengths at issue are the same order of magnitude as those experienced by users of commonly used appliances and devices such as electric blankets, so attempts to limit exposure would require significant changes in lifestyle for large segments of the population and significant design changes for many appliances, home wiring, and so on. In any case, the duration of exposure for individuals in these studies exceeds the duration of exposure of MRI patients by many orders of magnitude, so it is safe to conclude that MRI exposures do not make a significant contribution to population-wide ELF exposure at the present time.

Radio-Frequency Electromagnetic Waves

Radio waves are capable of heating biologic materials. Heating occurs because the conductivity of tissues supports the transfer of energy from the radio wave to induced electrical currents in the patient. The energy dissipated in the patient from resistance to the induced currents leads ultimately to heating of tissues in the patient. This energy dissipation is described in terms of the specific absorption rate (SAR) in units of watts per kilogram. For MRI systems, the phenomenon of tissue heating in the patient is of concern during the sending part of the imaging cycle. The SAR from MRI depends upon the particular pulse sequence used (the shape of the pulses, fraction of time that power is applied, etc.). It also depends upon the frequency of the radio waves and the size, density, and electric conductivity of the patient. This parameter has been calculated for models of the human head and torso^{28–30} and has been measured for

In addition to deleterious effects, induced electrical currents have been associated with at least one beneficial effect. Current densities of the order of 20 mA/m^2 have been used clinically to stimulate union of bone fractures. However, there is some controversy regarding the efficacy of this technique.¹⁸

A reversible and apparently innocuous phenomenon related to induced current densities above 10 mA/m^2 is the magnetophosphene phenomenon. This phenomenon was discussed previously in relation to motion of the patient's head in a static magnetic field. Changes in the magnetic field over time have the same result when the patient's head is stationary. These phenomena have been associated with rates of change of the magnetic field below 2 tesla/sec at repetition rates of 15 magnetic field pulses per second,¹² well within the operating parameters of current MRI systems.

cases of clinical interest.^{31,32} The maximum temperature rise resulting from a given SAR may be estimated from

$$T = \frac{\text{SAR } t}{C_H} \quad (25-1)$$

where T is the temperature rise in degrees Celsius, SAR is the specific absorption rate in watts per kilogram, t is the duration of exposure in seconds, and C_H is the heat capacity of the sample in joules per kilogram-degree Celsius. A typical value of heat capacity for human soft tissue is 3470 J/(kg-°C). Equation (25-1) yields a maximum estimate of temperature rise because it does not take into account effects such as conduction (vascular flow through a volume), convection (mixing that spreads out the effects of heating in a liquid), or radiation (heat energy given off in the form of infrared radiation). These effects act to reduce heating in most clinical situations below the amount calculated by Eq. (25-1).

Example 25-1

Find the maximum temperature rise produced during an MRI session in which a SAR of 3 W/kg is applied for 5 minutes.

$$\begin{aligned} T &= \frac{\text{SAR } t}{C_H} \\ &= \frac{3 \text{ W/kg (5 min)(60 sec/min)}}{3470 \text{ J/(kg-°C)}} \\ &= 0.26^\circ\text{C} \end{aligned} \quad (25-1)$$

The watt, the unit for power, equals 1 J of energy per second—that is, the rate of energy deposition. Therefore, all units except degrees Celsius cancel in Eq. (25-1).

It is generally agreed that heating of tissues in healthy humans is not of concern when the rise in core body temperature is less than 1°C. Cell killing is associated with temperatures above 42°C, approximately 5°C above core body temperature. There is some disagreement as to the biological effects of heating between these extremes.

It is known that many of the body's autonomic response mechanisms are reversible.³³ These mechanisms include the neuroendocrine and cardiovascular systems. Thus elicitation of a biologic response is not necessarily evidence of deleterious biologic effects. In addition to an overall temperature rise, there is some concern that “hot spots” could occur. If, for example, pathways of electrical current become narrowed in tissue with little vascularity, local heating effects may occur, although global models would not predict it. The presence of foreign materials such as ferromagnetic prostheses complicates the estimate of local heating and is considered later in this chapter.

Federal guidelines for RF exposure in humans for noninvestigational purposes are given in Table 25-2. It should be noted that these guidelines are based predominantly upon models for RF-induced heating in biologic materials. At present, the FDA approves scanning protocols for commercially available MRI systems on an individual basis. Most systems in clinical use have software indications of estimated SAR and do not allow operation of the system if guidelines are exceeded.

Biologic effects of RF exposure that are not directly related to heating and that may occur at low power levels have been suggested on a theoretical basis and have been reviewed extensively.³⁴ One of the most widely studied nonthermal effects is the alteration of calcium ion binding to nerve cell surfaces. However, specific waveforms not routinely encountered in MRI are required to elicit this effect.

Epidemiologic studies of groups such as military and civilian personnel exposed to radar equipment, and U.S. embassy employees exposed to microwave surveillance devices, have failed to produce conclusive evidence of harm.³⁵

SITE PLANNING

As with all major imaging systems, the acquisition of an MRI unit involves analysis and preparation of the site. In the case of MRI, the major factors to consider are (1) the effects of ferromagnetic materials upon the homogeneity of the magnetic field in

the bore of the magnet, (2) the effects of the static magnetic field upon metal objects in the scan room, (3) the effects of the static magnetic field outside of the scan room, (4) RF interference with the MRI scan produced by equipment outside of the scan room, and (5) RF interference with external communications produced by the MRI unit.

Site features that may complicate an MRI installation such as insufficient space, proximity of sensitive electronic components, and so on, are hallmarks of typical hospitals. A conceptually simple alternative to retrofitting existing space is construction of new space physically separate from the main hospital site. Although a number of severe problems were encountered in early installations, a great deal of experience has been accumulated. Vendors of MRI systems provide site planning services, and a number of general references are now available.³⁶ With foresight and planning, conversion of existing facilities is usually accommodated. Some of the major features of site selection and preparation are discussed in this section.

Magnetic Field Homogeneity

The presence of ferromagnetic materials in the vicinity of an MRI system may affect magnetic field homogeneity. Homogeneity is adjusted through the use of shim coils, electromagnets usually mounted on the ends of the main magnet bore that can be moved slightly with positioning screws to produce small variations in magnetic field strength. The effects of moving ferromagnetic materials such as elevators or laundry transport systems cannot be canceled by a static shimming system. At present there is no system to dynamically adjust magnetic field homogeneity. Therefore, the main system magnet should be located a distance away from large moving metallic objects. The minimum tolerable distance depends upon the requirements for magnetic field homogeneity. These requirements are more severe for large-bore, high-field strength magnets.

Radio-Frequency Interference

The presence of sources of RF energy external to the MRI system can interfere with MR signals from the patient and compromise the raw data accumulated during a scan. When the image is reconstructed, artifacts may occur because the amplitude of MR signals is increased at the frequencies of the interfering signals. When the image is reconstructed by a technique such as the Fourier transform, the interfering signals appear on the image as bands or other regular patterns. Some interference with the signals sent into the patient may also occur, but this effect is smaller because the magnitude of the signals sent into the patient is much greater than the magnitude of the signals received from the patient.

Sources of RF interference or electronic noise commonly found in the hospital include fluorescent lights, power supplies, electrical power lines, sprinkler systems, and communications networks. The connections between the MR scanner and the computer, RF power supplies, and gradient power supplies may also be significant sources of electronic noise.

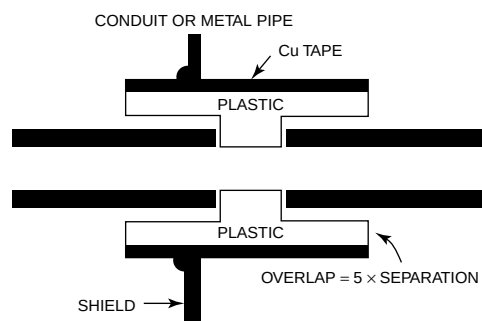
The basic plan of RF shielding is to surround the magnet with an electrically conductive enclosure. The enclosure isolates the scanner from unwanted signals conducted through metallic objects and from waves radiated through space. Continuity of the enclosure ensures a pathway for conduction of unwanted electric signals around the magnet so that they do not enter the bore and interfere with reception of signals from the patient. Penetration of RF waves into an enclosure is determined in part by the size of openings in the enclosure. Holes in the RF enclosure are generally kept smaller than one-half wavelength of the smallest interference wavelengths that can be tolerated. For example, the viewing window is usually covered with fine copper mesh screen having hole diameters of a millimeter or less.

When conduits or ventilation ducts must pass through the enclosure, waveguides are used to reduce unwanted RF penetration. A waveguide is constructed by breaking the conduit, inserting a plastic sleeve, and covering the outside of the sleeve with

Electromagnetic field theory, pioneered by British scientist Michael Faraday in the 1800s, predicts that a conducting surface will not be penetrated by electromagnetic radiation so long as there are no holes that are larger than the wavelength of the radiation. This means that for the relatively long wavelength radiation (on the order of meters) of concern in magnetic resonance imaging, a cage with bars spaced no wider than a fraction of a meter apart will shield the MR scanner from unwanted interference. Such a cage, known as a Faraday cage, can reduce the level of radiofrequency interference by as much as 100 dB. Faraday cages are not usually required in MR site planning but may be necessary in areas of unusual radio-frequency interference. The principle of the Faraday cage explains why you are much safer during a thunderstorm when in a closed car than when standing outside or even in most buildings. This affect is often misattributed to insulation provided by the tires. Tires do not provide enough insulation to prevent the extremely high voltages associated with a lightning strike from producing electrical currents. The car is safer because it acts as a Faraday cage. The electromagnetic energy can strike the surface of the car, but will not penetrate to the interior.

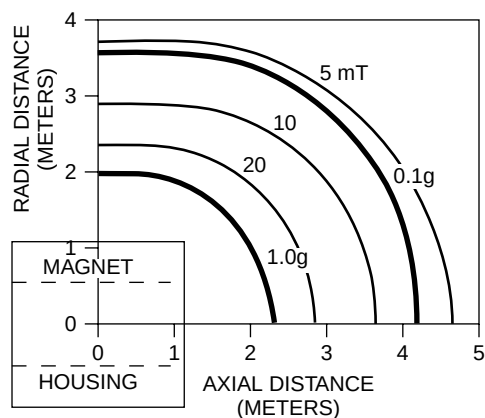
It is recommended that all patients undergo screening prior to admittance into the magnetic resonance scanner. The screening should check for the presence of foreign materials within the patient that may produce an adverse effect in the presence of static or changing magnetic fields. In addition, the FDA recommends particular caution with regard to the following patients¹¹:

- a. Patients with greater than normal potential for cardiac arrest.
- b. Patients who are likely to develop seizures or claustrophobic reactions.
- c. Decompensated cardiac patients, febrile patients, and patients with impaired ability to perspire.
- d. Patients who are unconscious, heavily sedated or confused, and with whom no reliable communication can be maintained.
- e. Examinations that are carried out at a room temperature above 24°C or at a relative humidity above 60%.

**MARGIN FIGURE 25-10**

Construction of a waveguide to allow conduit or a communication aperture for the MR scan room.

(From AAPM. *Site Planning for Magnetic Resonance Imaging Systems*, Report to AAPM NMR Task Group No. 2, Report No. 20. New York, American Institute of Physics, 1987. Used with permission.)

**MARGIN FIGURE 25-11**

Bold lines show the locations at which an iron object is pulled toward a 1.0-tesla MR system magnet with a force equal to the weight of the object (1.0 g) and with a force equal to one tenth the weight of the object (0.1 g). The magnet is assumed to contain no active shielding, and the iron is assumed to be in the form of a rod oriented along the lines of magnetic force, the shape and position that maximizes attraction. (From AAPM. *Site Planning for Magnetic Resonance Imaging Systems*, Report to AAPM NMR Task Group No. 2, Report No. 20. New York, American Institute of Physics, 1987. Used with permission.)

tape that is electrically conductive (Margin Figure 25-10). A small gap is maintained between the conductive tape and the conduit so that there is no continuous path for conduction of electricity along the conduit.

Patient and Personnel Safety

The presence of a high-strength magnetic field imposes limitations on the use of MRI and restrictions on the immediate surroundings of the scanner. Metal objects are attracted to the magnet of an MRI unit. The force with which they are attracted depends upon the type of material, the total mass, and the shape of the object. For example, rod-shaped objects aligned with the magnetic field lines are attracted with greater force than are spherical objects of the same material. At a distance d of several times the size of the magnet, the force varies approximately as $1/d^3$. The force is quite sensitive to small changes in distance, so attempts to “test” the pull of the magnet by moving an object close to the magnet may result in an unexpectedly sudden onset of a substantial force. One benchmark for safety is the distance at which the pull of the magnet overcomes the pull of gravity and renders the object weightless (Margin Figure 25-11). The pull of the earth’s magnetic field, approximately 0.05 millitesla, is small when compared with the gravitational attraction for objects of appreciable mass.

Patients must be examined for the presence of ferromagnetic objects such as surgical clips, shrapnel, prostheses, and so on. The three issues of concern for such objects are (1) torque or force upon the object within the patient, (2) heating of the material, and (3) distortion of the image. The presence of ferromagnetic objects within the patient may not contraindicate MRI. However, it is standard practice to interview the patient prior to a scan so that decisions may be made. For example, most institutions have established protocols for exclusion of patients who have ferromagnetic materials in the eye. In patient populations such as sheet metal workers or combat veterans who are suspected to be at risk, these protocols may include pre-MRI x rays to detect metallic objects.

Implants such as surgical clips, even though they may be made from stainless steel, may not be affected by magnetic fields because there are several different types of surgical-grade steel.^{37–41} The size and location of the clips are also important. In the absence of specific information, the most prudent course is to assume that the implant materials will be affected.

Prostheses such as plates in the skull, hip replacements, and other types of hardware that are firmly attached to bone are of little concern with respect to torque. However, they are of greater concern with respect to RF heating. As with direct RF heating of tissues, a temperature rise of less than 1°C for objects within the body is of little concern. The degree of heating depends upon the pulse sequence used, and is influenced by the mass of the implant. For example, at field strengths below 2 tesla, heating of single hip replacements is not usually significant, but double hip replacements may be of concern.⁴²

At the present time, the presence of a cardiac pacemaker mandates that the patient be excluded from MRI.^{43,44} Torque on the pacemaker is usually unacceptable within the bore of the magnet at moderate field strengths. Moreover, the RF fields and gradient switching can act to divert the programming circuitry of the pacemaker into an asynchronous mode. Therefore, individuals with pacemakers are usually excluded even from the control area of the scanner.

SUMMARY

- The main components of an MRI system are
 - Main system (static field) magnet
 - Gradient coils
 - Radio-frequency coils

- Transmitter
- Mixer
- Analog-to-digital converter
- Computer
- MRI system magnets that produce the static magnetic field are categorized as
 - Electromagnets
 - Resistive
 - Superconducting
 - Permanent
- MRI artifacts include
 - Foreign objects
 - Magnetic field inhomogeneities
 - Gradient nonlinearity
 - Radio-frequency interference
 - Motion
 - Chemical shift
- Quality assurance tests for MRI systems include
 - Daily tests
 - Resonance frequency
 - Signal-to-noise
 - Monthly tests
 - Image uniformity
 - Linearity
 - Hard copy
 - Yearly tests
 - Magnetic field homogeneity
 - RF pulse shape and amplitude
 - Gradient pulse shape and amplitude
- Bioeffects that are possible in MRI include
 - Movement of ferromagnetic objects into the magnet from the external environment
 - Movement of metallic devices, foreign bodies, or implants within the patient
 - Disruption of implanted devices
 - RF heating
 - Induction of electrical currents within the patient
- In site planning for MRI systems, the following must be considered:
 - Exclusion of metallic objects from the scan room
 - RF interference from the outside world and from the scanner
 - Cryogen ventilation in the event of a quench of a superconducting system
 - Methods to control traffic (i.e., excluding individuals wearing pacemakers)

REFERENCES

1. Hoult, D. I., and Lauterbur, P. C. The sensitivity of the zeugmatographic experiment involving human samples. *J. Magn. Reson.* 1979; **34**:425.
2. Crooks, L. E., Arakawa, M., Hoenninger, J., et al. Magnetic resonance imaging: Effects of magnetic field strength. *Radiology* 1984; **151**:127.
3. Ugurbil, K., Garwood, M., Rath, A. R., et al. Amplitude and frequency/phase modulated refocusing pulses that induce plane rotations even in the presence of inhomogeneous fields. *J. Magn. Reson.* 1988; **78**:472.
4. Hayes, C. E., Edelstein, W. A., and Schenck, J. F. Radiofrequency coils, in Thomas, S. R., and Dixon, R. L. (eds.), *NMR in Medicine, American Association of Physicists in Medicine, Medical Physics Monograph No. 14*. New York, American Institute of Physics, 1986, pp. 142–165.
5. Hyde, J. S., Froncisz, W., Jesmanowicz, A., et al. Planar-pair local coils for high-resolution magnetic imaging, particularly of the temporomandibular joint. *Med. Phys.* 1986; **13**:1–7.
6. Kneeland, J. B. Instrumentation, in Stark, D. D., and Bradley, W. G. (eds.), *Magnetic Resonance Imaging*. St. Louis, Mosby-Year Book, 1988, pp. 56–65.
7. Bradley, W. G., Waluch, V., Lai, K. S., et al. The appearance of rapidly flowing blood on magnetic resonance images. *AJR* 1984; **143**:1167–1174.
8. Knowles, R. J., and Markisz, J. A. *Quality Assurance and Image Artifacts in Magnetic Resonance Imaging*. Boston, Little, Brown & Co, 1988.
9. Dixon, R. L. *MRI: Acceptance Testing and Quality Control*, Proceedings of an AAPM Symposium, April 6–8, 1988, Bowman Gray School of Medicine. Madison, WI, Medical Physics Publishing Corp., 1988.
10. American College of Medical Physics. *Radiation Control and Quality Assurance Surveys: Magnetic Resonance Imaging, A Suggested Protocol*, Report No. 5. New York, American College of Medical Physics, 1989.
11. Guidance for the submission of premarket notifications for Magnetic Resonance Diagnostic Devices, Food and Drug Administration, Center for Devices

- and Radiologic Health, Computed Imaging Devices Branch, Office of Device Evaluation, FDA, Rockville, MD, www.fda.gov/chrh/ode/mri340.pdf, November 14, 1998.
12. Budinger, T. F., Cullander, C., and Bordow, R. Switched magnetic field thresholds for the induction of magnetophosphores. Presented at the Third Annual Meeting of the Society of Magnetic Resonance in Medicine, New York, August 13–17, 1984.
 13. Marsh, J. L., Armstrong, T. J., Jacobson, A. P., et al. Health effect of occupational exposure to steady magnetic fields. *Am. Indust. Hyg. Assoc. J.* 1982; **43**:387.
 14. Budinger, T. F., Bristol, K. S., Yen C. K., et al. Biological effects of static magnetic fields. Presented at the Third Annual Meeting of the Society of Magnetic Resonance in Medicine, New York, August 13–17, 1984.
 15. Milham, S. Mortality from leukemia in workers exposed to electrical and magnetic fields. *N. Engl. J. Med.* 1982; **307**:249.
 16. Rockette, H. E., and Arena, V. C. Mortality studies of aluminum reduction plant workers: Potroom and carbon department. *J. Occup. Med.* 1983; **25**:549.
 17. Tenforde, T. S., and Budinger, T. F. Biological effects and physical safety aspects of NMR imaging and *in vivo* spectroscopy, in Thomas, S. R., and Dixon, R. L. (eds.), *NMR in Medicine: The Instrumentation and Clinical Applications*, Medical Physics Monograph No. 14. American Institute of Physics, 1986, pp. 493–548.
 18. Bassett, C. A. L. Pulsing electromagnetic fields: A new approach to surgical problems, in Buchwald, H., and Varco, R. L. (eds.), *Metabolic Surgery*. New York, Grune & Stratton, 1978, pp. 255–306.
 19. Luben, R. A., Cain, C. D., Chen, M. C.-Y., et al. Effects of electromagnetic stimuli on bone and bone cells *in vitro*: Inhibition of responses to parathyroid hormone by low-energy low-frequency fields. *Proc. Natl. Acad. Sci. USA* 1982; **79**:4180.
 20. Hoult, D. I., and Lauterbur, P. C. The sensitivity of the zeugmatographic experiment involving human samples. *J. Magn. Reson.* 1979; **34**:425–433.
 21. Weaver, J. C., and Astumian, R. D. The response of living cells to very weak electric fields: The thermal noise limit. *Science* 1990; **247**:459–462.
 22. Roberts, N. J., Jr., Michaelson, S. M., and Lu, S. T. The biological effects of radiofrequency radiation: A critical review and recommendations. *Int. J. Radiat. Biol.* 1986; **50**:379–420.
 23. McDowall, M. E. Mortality of persons resident in the vicinity of electricity transmission lines. *Br. J. Cancer* 1986; **53**:271–279.
 24. Skeikh, K. Exposure to electromagnetic fields and the risk of leukemia. *Arch. Environ. Health* 1986; **41**:56–63.
 25. Easterly, C. E. Cancer linked to magnetic field exposure: A hypothesis. *Am. J. Epidemiol.* 1981; **114**:169–174.
 26. Savitz, D. A., and Calle, E. E. Leukemia and occupational exposure to electromagnetic fields: Review of epidemiologic surveys. *J. Occup. Med.* 1987; **29**:47–51.
 27. Elder, J. A. Radiofrequency radiation activities and issues: A 1986 perspective. *Health Phys.* 1987; **53**:607–611.
 28. Brown, R. W., Haacke, E. M., Martens, M. A., et al. A layer model for RF penetration, heating, and screening in NMR. *J. Magn. Reson.* 1988; **80**:225–247.
 29. Bottomley, P. A., and Andrew, E. R. RF magnetic field penetration, phase shift and power dissipation in biological tissue: Implications for NMR imaging. *Phys. Med. Biol.* 1978; **23**:630–643.
 30. Bottomley, P. A., Redington, R. W., Edelstein, W. A., et al. Estimating radio-frequency power deposition in body NMR imaging. *Magn. Reson. Med.* 1985; **2**:336–349.
 31. Roschmann P. Radiofrequency penetration and absorption in the human body: Limitations to high-field whole body nuclear magnetic resonance imaging. *Med. Phys.* 1987; **14**:922–932.
 32. Shuman, W. P., Haynor, D. R., Guy, A. W., et al. Superficial and deep-tissue temperature increases in anesthetized dogs during exposure to high specific absorption rates in a 1.5T MR imager. *Radiology* 1988; **167**:551–554.
 33. Parker, L. N. Thyroid suppression and adrenomedullary activation by low-intensity microwave radiation. *Am. J. Physiol.* 1973; **224**:1388.
 34. Postow, E., and Swicord, M. L. Modulated Field and “Window” Effects, in Polk, C., and Postow, E. (eds.), *Handbook of Biological Effects of Electromagnetic Fields*. Boca Raton, FL, CRC Press, 1986.
 35. Tenforde, T. S., and Budinger, T. F. Biological Effects and Physical Safety Aspects of NMR Imaging and *in Vivo* Spectroscopy, in Thomas, S. R., and Dixon, R. L. (eds.), *NMR in Medicine: The Instrumentation and Clinical Applications*, Medical Physics Monograph No. 14. New York, American Institute of Physics, 1986, pp. 521–522.
 36. AAPM. *Site Planning for Magnetic Resonance Imaging Systems*, Report of AAPM NMR Task Group No. 2, AAPM Report No. 20. New York, American institute of Physics, 1987.
See also MR equipment vendors (e.g., Siemens Medical Systems Inc., Iselin, NJ; General Electric, Milwaukee, WI; Picker International, Cleveland, Ohio; Philips Medical Systems, Shelton, CT; Elscint Ltd., Boston, MA).
 37. Barrafato, D., and Helkelman, M. Magnetic resonance imaging and surgical clips. *Can. J. Surg.* 1984; **27**:509.
 38. Dujovny, M., Kossowski, N., Kossowsky, R., et al. Aneurysm clip motion during magnetic resonance imaging: In vivo experimental study with metallurgical factor analysis. *Neurosurgery* 1985; **17**:543–548.
 39. Finn, E. J., DeChiro, G., Brooks, R. A., et al. Ferromagnetic materials in patients: Detection before MR imaging. *Radiology* 1985; **156**:139–141.
 40. Laakman, R. W., Kaufman, B., Han, J. S., et al. MR imaging in patients with metallic implants. *Radiology* 1985; **157**:711–714.
 41. New, P. F. J., Rosen, B. R., Brady, T. J., et al. Potential hazards and artifacts of ferromagnetic and nonferromagnetic surgical and dental materials and devices in nuclear magnetic resonance imaging. *Radiology* 1983; **147**:139–148.
 42. Davis, P. L., Crooks, L., Arakawa, M., et al. Potential hazards in NMR imaging: Heating effects of changing magnetic fields on small metallic implants. *AJR* 1981; **137**:857–860.
 43. Pavlicek, W., Geisinger, M., Castle, L., et al. The effects of nuclear magnetic resonance on patients with cardiac pacemakers. *Radiology* 1983; **157**:149–153.
 44. Erlebacher, J. A., Cahill, P. T., Pannizzo, F., et al. Effect of magnetic resonance imaging on DDD pacemakers. *Am. J. Cardiol.* 1986; **57**:437–440.

EXPERIMENTAL RADIOBIOLOGY

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OBJECTIVES

After studying this chapter, the reader should be able to:

- Describe models for the direct and indirect effects of radiation.
- Characterize the components of a cell survival curve.
- Explain the effects of oxygen, LET, dose, dose rate, and fractionation on cell survival studies.
- Provide examples and proposed mechanisms of action of representative radio-protecting and radiosensitizing agents.
- Distinguish between somatic and genetic effects of radiation.

INTRODUCTION

An “almost instantaneous” interaction is one that occurs in less than 10^{-18} seconds.

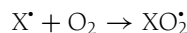
A free radical is an electrically neutral chemical species that has an unpaired electron and is chemically very reactive. Electrons possess the quality “spin,” and electrons tend to pair together, with one spin “up” and one spin “down.” The presence of an unpaired electron in a free radical is an unstable situation that is remedied when the free radical either acquires or shares an electron from another molecule. Thus free radicals act as strong oxidizing agents and disrupt chemical bonds in the process.

Direct effects of radiation are caused by electrons released during interactions of radiation directly in biological material. Indirect effects of radiation are produced by free radicals formed as radiation interacts in biological material.

For direct effects, the relationship between dose and number of entities (molecules, cells, etc.) affected is expressed as $E = E_0(1 - e^{-kD})$, where E is the number affected, E_0 is the number of entities present, D is the absorbed dose, and k is a constant called the “inactivation constant.” The value $1/k$ is the dose required to leave only 37% ($1/e$) of the entities unaffected. When the entities are cells, $1/k$ is termed the *mean lethal dose*.

Living organisms contain large amounts of water in both free and bound molecular states. Water plays a central role in radiation interactions within biological systems. During these interactions, water molecules are ionized by removal of electrons, leaving positively charged molecules (H_2O^+). The released electrons combine almost instantaneously with neutral water molecules to form negatively charged molecules (H_2O^-). Almost instantaneously means less than 10^{-18} sec, the shortest measurable time interval. The positively and negatively charged water molecules are unstable, dissociating into ions (H^+ and OH^-) and *free radicals* ($H^\bullet$ and $OH^\bullet$).

Free radicals in biologic specimens tend to react within 10^{-5} sec. They react among themselves and with water, organic molecules, oxygen, and their own reaction products. The products of free radical reactions include poisons such as hydrogen peroxide (H_2O_2). Of particular interest is the interaction of free radicals with oxygen, which serves to prolong the presence of free radicals. When a free radical $X^\bullet$ interacts with molecular oxygen O_2 , the product is a longer-lived free radical.



Oxygen serves to heighten the effects of free radicals by allowing the radicals to “survive” long enough to migrate farther from their origin, thus expanding the potential for biologic damage.

Because they are produced at one or more steps removed from the initial ionizing event, radiobiologic effects of free radicals are termed *indirect effects*. Effects produced directly by electrons set in motion as radiation interacts in a material, such as disruption of chemical bonds in macromolecules, are termed direct effects. These effects are included in the summary of radiation interactions depicted in Figure 26-1.

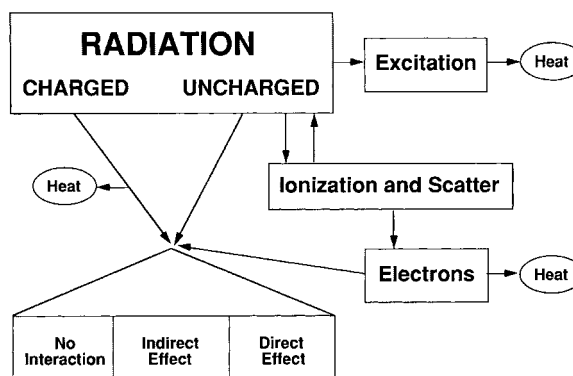


FIGURE 26-1

As radiation (charged particles or uncharged particles such as photons) enters matter, various outcomes are possible, including no interaction, an indirect effect (free radical production), a direct effect (disruption of chemical bonds in biomolecules), and the deposition of heat in the material.

■ INTERACTIONS AT THE CELL AND TISSUE LEVELS

Radiation is energy in motion. Radiation damage is the consequence of transfer of sufficient energy from radiation to tissue to produce a biological effect. The mechanism of damage is important. Although heat is always produced during radiation interactions, the mechanism for radiation bioeffects is definitely not heating of tissue. Calculation shows that a dose of over 4 kGy is required to raise the average temperature of water (or soft tissue) by 1°C. If heating were the mechanism of radiation bioeffects, the effects would be negligible below a dose of several kilogray, because temperature elevations on the order of 1°C or less are overshadowed by normal metabolic fluctuations in mammalian organisms.

The effects of radiation are traceable to the disruption of chemical bonds. Because several repair mechanisms exist for important macromolecules and also because the concentration of molecules often exceeds the minimum concentration required for cellular function, massive amounts of chemical disruption would be necessary to cause discernible effects. There is, however, an exception to this conclusion, because there exists a critical site of radiation damage within cells. That site is DNA.

The impact of radiation on DNA affects the ability of the cell to function and reproduce. Alteration of most biological molecules simply reduces the number of those molecules available for chemical reactions. However, alteration of DNA causes information to be “read” erroneously by the cell so that it loses or changes its normal function. If the cell in question is a germ cell, the transfer of information to the next generation of the organism may be compromised.

Ionizing radiation is particularly effective in disrupting DNA because it can produce damage in DNA components (the base, phosphate, and deoxyribose fractions), all within a relatively small region.

DNA is a critical site for direct radiation damage in a cell because copies of the DNA molecule are not present.

Through its ability to produce local, multiply damaged sites in DNA, ionizing radiation can induce the same amount of cell killing as aflatoxin, benzopyrene, and other agents known to create DNA damage.¹

■ CELL SURVIVAL STUDIES

Cells may be grown in culture, surviving on nutrients provided by growth media and replicating on a time scale that reflects the growth kinetics of the specific cell line. The effects of various agents on cell survival under controlled laboratory conditions can be studied in these cell cultures. Direct observation of metabolic processes in cells is difficult; a more practical end point for observation is cell death (or survival). However, the definition and detection of cell death is also challenging. A more easily recognized end point is loss of reproductive capacity. After exposure to an agent, cell survival can be evaluated in terms of the fraction of cells that are able to replicate as evidenced by their ability to form colonies in culture.

A graph showing the surviving fraction (defined in terms of reproductive capacity) of a cell population as a function of dose of an agent is referred to as a *cell survival curve*. It is conventional to show the surviving fraction on a log scale. Most cell survival curves have at least a portion that is a straight line, which suggests an exponential equation to describe the curve.

A typical cell survival curve for x rays is shown in Figure 26-2. By definition, the fraction of surviving cells is unity when the dose is zero. As the dose is increased, the surviving fraction may remain equal to or near 1. Small doses of an agent may have no effect on reproductive capacity; or if cells are damaged, the damage may be repaired prior to the next replication. This *repair of sublethal damage* contributes to the shoulder observed on cell survival curves for x rays.

Beyond the shoulder, the cell survival curve displays a linear portion. Here the surviving fraction N/N_0 is described by the equation

$$\frac{N}{N_0} = e^{-D/D_0} \quad (26-1)$$

When the dose D equals D_0 , the mean lethal dose, the surviving fraction is e^{-1} , or 0.37.

The technique of using cultured cells to study radiation effects was developed in the 1950s by Puck and his colleagues at the University of Colorado.²

The inability of a cell to divide and form a colony is referred to as “reproductive death.” A cell that experiences reproductive death may still be “metabolically alive.”

The exponential nature of cell killing is reminiscent mathematically of radioactive decay, the attenuation of x rays, and the relaxation of nuclear spins.

The mean lethal dose is defined as the dose required to reduce the “survival” to e^{-1} or 37% along the straight line portion of the survival curve. D_0 is 1 to 2 Gy for most mammalian cells exposed to x rays.

Douglas Lea was one of the pioneers in mathematical modeling of cellular radiation injury. His book *Actions of Radiation on Living Cells* (Cambridge University Press) is a classic in the radiation biology literature.

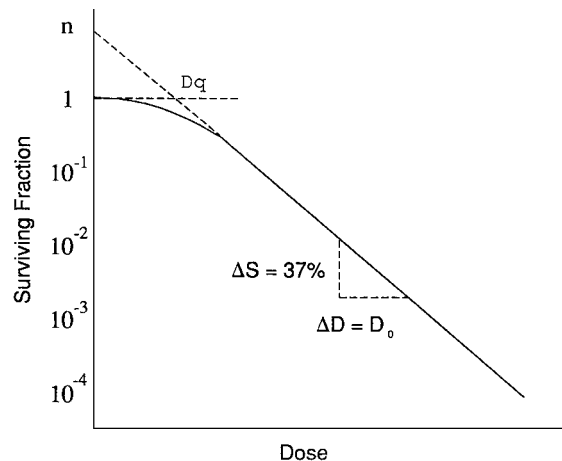


FIGURE 26-2

Representation of a cell survival curve illustrating extrapolation number n , quasithreshold dose D_q , and mean lethal dose, D_0 .

The target theory of radiation interactions in cells was developed in the 1950s by Lea.³

Mathematical models of radiation-induced loss of cellular reproductive capacity contain several assumptions:

- Loss of reproductive capacity is a multistep process.
- Absorption of energy in a critical volume(s) of the cell is the first step.
- Energy absorption creates molecular lesions in the cell.
- These lesions cause the loss of the cell's ability to carry out normal DNA replication and cell division.⁶

Three types (lines) of mammalian cells have been widely employed in tissue-culture studies of radiation effects. These lines are HeLa cells derived from a patient with cervical carcinoma, CHO cells derived from a Chinese hamster ovary, and T1 cells with a human kidney origin.

The width of the shoulder on a cell survival curve also reflects the number of sensitive targets that must be inactivated in the cell before the cell survival curve begins to decrease exponentially (i.e., reach the straight-line portion of a semilog plot). A “single-hit, multitarget” model of cell survival implies that there is no repair of sublethal damage and that only a single damaging “event” is necessary to inactivate each sensitive target within the cell. The number of targets that must be inactivated to kill the cell is found by extrapolating the straight-line portion of the cell survival curve back to the vertical axis. The value of the surviving fraction at the intersection with the vertical axis is the *extrapolation number* n , representing the average number of sensitive targets per cell. A wider shoulder yields a point of intersection higher on the vertical axis, implying that more targets must be inactivated. The extrapolation number n for mammalian cells exposed to x rays is highly variable and ranges from 1 (i.e., no shoulder, one target per cell) up to 20. The variability reflects the error inherent in extrapolation as well as the variability in cell types.

In actual cell survival curves, the shoulder probably indicates the effects of repair enzymes as well as the number of targets. It may not be possible to identify and specify the characteristics of sensitive targets within cells by using survival curves alone.⁴ Models that are more complex than the multitarget single-hit model have been proposed.⁵

Another measure of the shoulder of a cell survival curve is the *quasithreshold dose* D_q . If the surviving fraction of cells is unity up to a certain dose and decreases exponentially beyond that dose, then such a dose would be a true threshold, a dose below which there is no cell killing. The quasithreshold dose, while not implying a true threshold, characterizes the end of the shoulder. It may be estimated as the dose at which a horizontal line, at the level of a surviving fraction of 1, crosses the extrapolated straight-line portion of the cell survival curve. It may also be shown⁷ that

$$D_q = D_0 \ln(n)$$

Given the previous estimates for D_0 and n , D_q for mammalian cells ranges from 0 to 6 Gy.

■ MODIFICATION OF CELLULAR RESPONSES

Factors that influence the effects of radiation upon cells may be characterized by their effects upon cell survival curves. These factors include the presence of oxygen, linear energy transfer (LET) of the radiation, dose rate, fractionation, phase of the cell cycle, and presence of radiation modifiers.

Oxygen

Oxygen influences the indirect effects of ionizing radiation by modifying free radical interactions. Hence, the partial pressure of oxygen in the vicinity of the cell modifies the cell sensitivity. Two cell populations with different partial pressures of oxygen are illustrated in the margin. Population A is fully oxygenated and population B is less oxygenated. At a given dose beyond the quasithreshold dose, the surviving fraction is lower for population A. For a specific biologic effect at a particular partial pressure of oxygen, the oxygen enhancement ratio (OER) is defined as

$$\text{OER} = \frac{\text{Dose required with no oxygen present}}{\text{Dose required at given partial pressure of oxygen}}$$

The OER does not exceed 3 for mammalian cells and increases little for partial pressures above 1 atm.

Linear Energy Transfer

LET is a measure of the density of ionizing events in the medium. As the LET is increased, the width of the shoulder of cell survival curves tends to decrease, reflecting an increase in damage per unit dose. Types of ionizing radiation that deposit more energy over shorter spatial intervals have a higher probability of damaging DNA within the cell, compared with radiations in which ionization is spread over larger distances. Therefore, damage is increased for a given dose if LET is increased.

The width of a single strand of DNA is on the order of a few thousandths of a micrometer. Neutrons, having an LET of 20 keV/μm, exhibit a higher probability of interacting more than once in the thickness of a DNA molecule, compared with x rays with an LET of less than 1 keV/μm. Cell populations exposed to low- and high-LET radiation are compared in the margin.

Higher-LET radiation produces relatively more damage by direct effects than by indirect effects. Thus, the oxygen effect, basically a result of indirect effects, is diminished for higher-LET radiation. Also, the disappearance of the shoulder for higher-LET radiation implies that effects related to repair of sublethal damage also diminish.

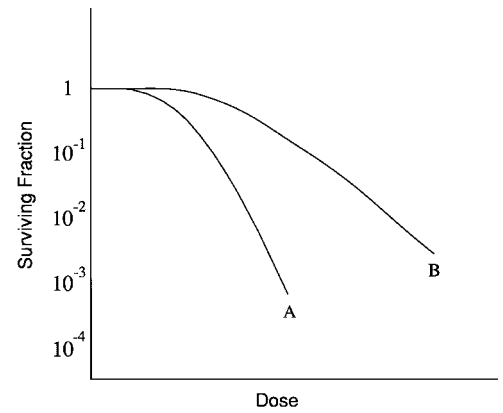
Dose Rate and Fractionation

Compared with radiation delivered at a higher dose rate, radiation delivered at a lower rate may produce less cell killing even though the total doses are identical. Also, radiation delivered at a high rate but in fractions or discrete doses separated in time may be less damaging to cells than the same amount of radiation delivered in a single session. Repair of sublethal damage is the mechanism underlying dose rate and fractionation effects. An increase in dose rate may produce lesions at a frequency that exceeds the capacity of the cell for repair. The aggregate amount of damage produced in a short, high dose-rate fraction may be repairable if a suitable interval is allowed before additional irradiation. The effect of increasing dose rate upon cell survival is shown in the margin.

In Margin Figure 26-4, the cell survival curve for treatments given in a single dose is shown as curve abc, while the curve for an identical cell population in which a recovery period is allowed is curve abd. A recovery period of 10 to 20 hours is required for a detectable modification in mammalian cell survival through fractionation. This minimum period between fractions reflects the timing of events in the mammalian cell cycle.

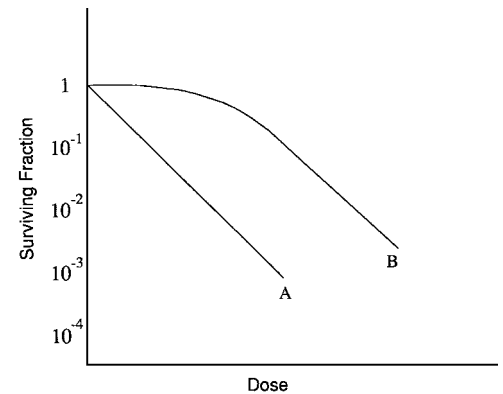
Cell Cycle

Figure 26-3 shows a typical cell cycle for somatic mammalian cells grown in culture. The cell cycle is divided into two parts, interphase and mitosis. During interphase, which occupies the majority of time in the cell cycle for somatic cells, the amount of cellular DNA is doubled as replicas of the chromosomes are synthesized. Interphase



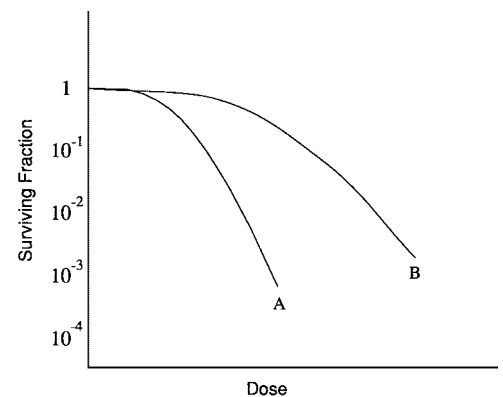
MARGIN FIGURE 26-1

Cell survival curves showing the effect of oxygenation. The cell population represented by curve A is fully oxygenated, with population B less oxygenated.



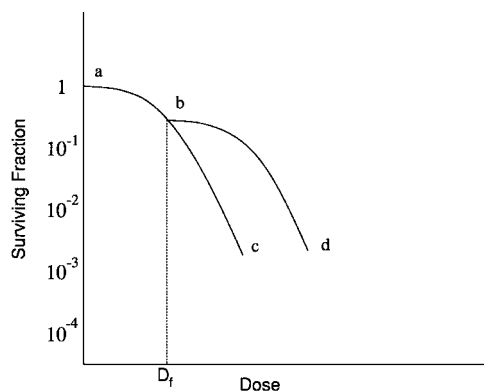
MARGIN FIGURE 26-2

Cell survival curves showing the effect of linear energy transfer (LET). The LET of the radiation is lower for B than for A. The shoulder disappears as LET increases.



MARGIN FIGURE 26-3

Cell survival curves showing the effect of dose rate. The radiation is delivered at a higher dose rate for A than for B.

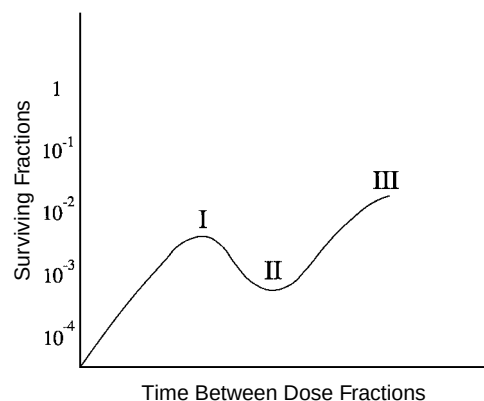


MARGIN FIGURE 26-4

Cell survival curves showing the effect of fractionated treatment. Cell populations were irradiated in single doses at all points along curve abc. To construct curve abd a recovery period of 10 to 20 hours was allowed for any dose greater than D_r , the total dose fraction.

The mitotic spindle is a framework for coordinated movement of the chromosomes.

The radiation sensitivity of cells varies in different stages of the cell cycle. In general, mammalian cells are most resistant in late S phase and most sensitive in late G₂ and mitosis.



MARGIN FIGURE 26-5

A typical split-dose survival curve showing cell survival as a function of time allowed between doses. Stages I, II, and III are discussed in the text.

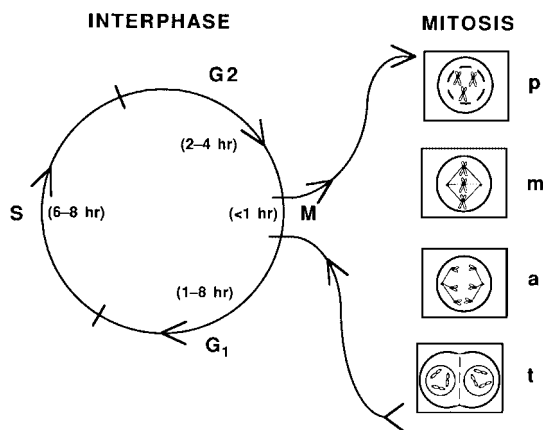


FIGURE 26-3

The cell cycle in mammalian cells is divided into interphase and mitosis. During interphase, DNA synthesis S is separated from mitosis M by gaps G₁ and G₂ during which normal activity of the cell is maintained. In mitosis or fission the nuclear membrane disappears in prophase p, a mitotic spindle forms in metaphase m, chromosomes separate in anaphase a, and nuclear and cellular membranes reappear to complete the separation of two progeny cells in telophase t.

is subdivided into three stages: G₁, or first gap, lasting 1 to 8 hours; S or DNA synthesis, lasting 6 to 8 hours; and G₂, or second gap, lasting 2 to 4 hours. During mitosis, the actions leading to cell division (fission of the cell) take place during a time period of approximately 1 hour. These actions are classified into four stages. During the first stage of mitosis, *prophase*, the nuclear membrane disappears. In the second stage, *metaphase*, a mitotic spindle forms, and the chromosomes become attached to the spindle. During the third phase, *anaphase*, the chromosomes migrate along the spindle to opposite sides of the cell. During the last stage, *telophase*, nuclear and cell membranes reappear, and two cells are present where formerly there was only one. The length of time spent by particular cells in each stage of the cell cycle, and the total length of the cell cycle, affect the radiation sensitivity of the cell.⁸

Split-Dose Survival Curve

In general, radiation is less damaging to cells if given in discrete doses or fractions rather than all at once. Fractionation techniques are used in radiation therapy to provide normal cells time for recovery. Tumor cells often lack many of the functional capabilities of the cells from which they are derived. When the repair capabilities of tumor cells are below those of normal cells, fractionated radiation therapy treatments give normal cells an advantage in repairing sublethal damage. Therefore, fractionation tends to increase the ratio of damage to tumor cells compared with damage to normal cells.^{9,10} A split-dose survival curve is shown in the margin. The shape of this curve illustrates many of the radiobiology principles discussed in this section. An initial dose fraction is given at time zero. The curve illustrates the effect upon survival when the second dose fraction is given at different times after the first fraction. At the time of the initial dose fraction, radiation encounters cells in all stages of division. Because the sensitivity of cells varies with the stage of the cell cycle, the cells that survive the first dose fraction are predominantly those that happened, by chance, to be in the radioresistant S phase. Thus, the second dose fraction encounters a cell population that is not randomly distributed. Stage I in the figure represents repair of sublethal damage between the dose fractions. Stage II represents the effect of waiting to administer the second dose fraction until the cells that were in S phase after the first dose fraction have progressed into the more sensitive G₂ and M stages. Stage III represents the effect of repopulation; after the length of a cell cycle, the survivors of the first dose fraction divide. Thus the surviving fraction actually increases during this stage because new cells are produced between fractions.

Radioprotectors and Radiosensitizers

Chemical agents may be used to modify the response of cells to ionizing radiation. These agents generally influence the indirect effects of radiation—that is, the effects of free radicals. Because of the speed with which free radical damage is produced, the agents must be present during irradiation. It is more difficult to modify the direct effects of ionization. Since the direct effects of radiation dominate as LET is increased, high-LET radiation effects are usually not modified significantly by administration of chemical agents.

Agents that reduce the effects of radiation upon cells are termed *radioprotectors*. Substantial interest in radioprotectors has arisen because of their potential applications in military and space programs as well as for medical uses such as protection of normal tissues during radiation therapy. One group of researchers has synthesized over 2000 potentially radioprotective compounds in the last 40 years.¹¹ Toxicity, unproven effectiveness of radioprotectors at low doses, and a lack of evidence for bioeffects of ionizing radiation at low doses make the use of radioprotectors in diagnostic radiology impractical.

Mechanisms of action for radioprotectors include inactivation of free radicals by preferential binding or scavenging, induction of hypoxia to reduce the sensitizing presence of oxygen, and occupation of sensitive sites in biomolecules by forming covalent bonds that are less easily disrupted by free radicals. Some protection against both direct and indirect effects has also been achieved by blocking cells in the G₁, phase of the cell cycle, thereby allowing more time for the action of repair enzymes before the S phase where DNA is replicated. The efficacy of radioprotectors is measured by the dose-modifying factor (DMF), where

$$\text{DMF} = \frac{\text{Dose to produce an effect with the radioprotector}}{\text{Dose to produce an effect without the radioprotector}}$$

DMFs do not usually exceed 2. The theoretical limit of a DMF is 3 because removal of all indirect effects of radiation still leaves approximately one-third of the damage due to direct effects. The DMF varies with cell type, stage of the cell cycle, and LET of the radiation.

Radiosensitizers are agents that increase the effects of radiation on cells. Radiosensitizers are most applicable in radiation therapy, where distinction must be made between “apparent” sensitizers and “true” sensitizers. Apparent radiosensitizers are actually toxic agents that are particularly effective in cases where radiation alone is less effective. True radiosensitizers such as analogues of pyrimidines (5-bromodeoxyuridine [5-BUDR] and 5-iododeoxyuridine [5-IUDR]) and purines (6-mercaptopurine) promote both direct and indirect effects of radiation by inhibiting repair enzymes and weakening the sugar–phosphate bond of the DNA molecule.

The direct effects of radiation may also be enhanced through the use of compounds containing boron to enhance the effects of neutron radiation therapy.¹² Low-energy neutrons (0.025 eV, or “thermal neutrons”) have an unusually high probability of interaction with boron compared with elements that are normally present in the body. When a neutron interacts with ¹⁰B, the unstable nuclide ¹¹B is formed. This nuclide fissions to produce α-particles that deliver a high absorbed dose in the immediate vicinity of the boron-containing compound. Although boron neutron capture therapy (BNCT) has been pursued experimentally over several decades, clinical applications have proven unsuccessful to date.

Sulfhydryl components such as cysteine and cysteamine are good free-radical scavengers.

Epinephrine and carbon monoxide are often used to induce hypoxia.

The amount of a radioprotector that can be administered is usually limited by its toxic effect on cells.

Examples of apparent radiosensitizers include: methotrexate and 5-fluorouracil (5-FU), which tend to kill cells in the radioresistant S phase; and electron-affinic compounds such as nitroimidazoles, which are toxic to hypoxic cells.

5-Bromodeoxyuridine (5-BUDR) and 5-iododeoxyuridine (5-IUDR) are radiosensitizing pyrimidine analogues, and 6-mercaptopurine is a radiosensitizing purine analogue.

BNCT as a method to enhance radiation effects has been revisited several times over the past few decades. Each of these renewed efforts has failed to date to produce a clinically useful approach to radiation therapy.

■ ANIMAL STUDIES

Knowledge of the basic interactions of radiation at the cellular level is the foundation for understanding the bioeffects of radiation exposure. However, predictions based upon simple cell models must be tested against actual exposure of organisms. Radiation effects involve complicating factors such as the presence of co-carcinogens and the actions of the immune system. These effects must be explored at a level of system

The word “somatic” is derived from the Greek *somatikos*, meaning “pertaining to the body (soma).”

The word “genetic” is derived from the Greek *gennan*, meaning “pertaining to the gene (gennan).”

The induction of cancer (carcinogenesis) is an often-studied end point for radiation effects in exposed animal populations. These studies have revealed three distinct stages of carcinogenesis: initiation, promotion, and progression.²⁰

William Bateson first used the word “genetics” in 1905 to define the discipline of variation and heredity.

Mutations in somatic cells are referred to as “somatic mutations.” Mutations in germ cells are referred to as “genetic mutations.”

Radiation-induced genetic effects in mice include

- Dominant lethal mutations
- Reciprocal gene translocations, sex chromosome losses and gains
- Recessive lethal mutations
- Recessive mutations causing morphological and biochemical traits
- Dominant mutations affecting histocompatibility genes
- dominant mutations causing
 - Skeletal abnormalities and cataracts
 - Prewaning mortality
 - Congenital abnormalities
 - Heritable tumors²⁵

complexity above that of the cell. Both somatic (from the Greek *soma*, “of the body”) and genetic effects are of interest.

Somatic Effects

Studies of animal survival following large doses of ionizing radiation have demonstrated a range of lethal doses that vary with the species and the irradiation conditions. A widely used measure of lethality is the dose required to kill 50% of the test animals within 30 days. This dose is called the LD_{50/30}. Among mammalian species the LD_{50/30} varies from about 3 to 7 Gy, with a trend toward lower LD_{50/30} for larger mammals. Death at these dose levels is caused by depletion of hematopoietic precursor cells, resulting in death by infection and bleeding due to a lack of mature erythrocytes, leukocytes, and platelets in the circulating blood supply. Larger species tend to have fewer hematopoietic system stem cells per unit body weight and are therefore more susceptible to these effects of radiation.^{11,13,14}

Biological end points other than death have been studied extensively in animal populations. Tumor systems have been examined with regard to the effects of radiation therapy alone or in combination with other agents.¹⁵ Normal tissue response has also been investigated in vivo by using end points such as skin reactions,¹⁶ breathing rate (for lung tissue),¹⁷ neurologic tests (for the spinal cord),¹⁸ and sterilization (for gonadal irradiation).¹⁹

Genetic Effects

In any animal population, there is a natural rate of spontaneous mutation. Because genetic variation is one of the factors that drives natural selection, it might be suggested that an increase in mutation rate could be beneficial to a population. However, there is no generally accepted evidence of this suggestion in experimental populations. In fact, it is generally assumed that in the absence of other selection pressures, an increase in the background mutation rate beyond a certain point would be deleterious.

One measure of the genetic impact of an agent is the doubling dose. This dose is the amount of an agent required to double the spontaneous mutation rate in a population. Experiments suggest that the spontaneous mutation rate in most mammalian species is doubled by an absorbed dose to the germ cells of 1 Gy.²¹ Some epidemiologic studies of human populations suggest that the doubling dose is slightly higher for humans.

Genetic effects have been studied extensively in a number of species. Dominant mutations may be demonstrated in first-generation offspring and identified within two generations, so studies may be carried out in a reasonable length of time in rodents and other populations that multiply rapidly.^{22,23} Longer-term multigenerational studies have been conducted since the 1920s in short-lived species such as the fruit fly, *Drosophila melanogaster*.^{21,24} By the early 1950s, there was considerable evidence that radiation-induced mutation rates were higher in rodents than in the fruit fly. Furthermore, it was found that mutation incidence in *Drosophila* varied linearly with dose and appeared to have no threshold. Moreover, there appeared to be no dose rate effect, suggesting that damage caused by exposure of germ cells is cumulative. If these effects occur in humans, then even a small amount of exposure per year, when summed over a long career, would lead to an appreciable risk of mutation.

To answer some of the questions raised by the *Drosophila* data, a series of large-scale experiments (involving millions of mice, dubbed the “mega-mouse” experiments) were carried out in the early 1950s. These experiments were designed to examine the genetic effects of ionizing radiation upon specific gene loci in mice. These loci determine observable traits such as coat color and ear size^{26–28} and can therefore be followed in experimental populations. One of the most important conclusions of the experiments was that for genetic effects, dose rate is an extremely important variable. At very low dose rates, no significant genetic effects were found, even when the total dose exceeded several gray. These results tempered the concern over occupational exposure of germ cells in humans where the dose rate is low.

Another result of the mega-mouse experiments was that delay of conception reduced the rate of genetic malformations in offspring. Thus, for gonadal doses large enough to create concern about genetic effects, (above 0.1 Gy to the gonads from radiation therapy or prolonged fluoroscopic procedures, for example), delay of conception for up to 6 months following exposure could be helpful in preventing genetic effects.¹¹

Japanese A-bomb survivors and their descendants have been studied for more than 50 years to identify radiation-induced genetic effects in humans. Of the eight indicators of genetic abnormalities monitored, none has shown a statistically significant increase in the exposed population compared with controls.²⁹

CONCLUSIONS

- Radiation acts directly on sensitive structures in cells (direct action) and through creation of sensitive intermediates (indirect action) to produce biologic damage.
- DNA (deoxyribonucleic acid) is a major sensitive site in the cell for the direct action of ionizing radiation.
- The loss of reproductive capacity of cells is a major way to describe cell "death" caused by ionizing radiation. This description is usually through the use of cell survival curves.
- Characteristics of a cell survival curve include the extrapolation number n , quasi-threshold dose D_q , and mean lethal dose D_0 .
- Factors that modify the radiation response of cells include the presence of oxygen, linear energy transfer (LET) of the radiation, total dose, dose rate, and dose fractionation.
- Mammalian cells are most sensitive when irradiated in the late G_2 and M phases of the cell cycle, and they are least sensitive in the S phase.
- Certain chemical substances can either heighten or suppress the response of cells to radiation. The action of these substances, known as radiosensitizing and radioprotecting agents, is described by the dose modifying factor (DMF).
- Actions of radiation directly on cells of the body other than germ cells are related to the somatic effects of radiation exposure.
- Actions of radiation on cells, giving rise to biological effects in future generations, are described as the genetic effects of radiation exposure.
- The doubling dose is the dose of ionizing radiation needed to double the spontaneous mutation rate in a biological organism.

REFERENCES

1. Ward, J. F., Limoli, C. L., Calabro-Jones, P., et al. Radiation vs. chemical damage to DNA, in Nygaard, D. F., Simic, M., and Cerutti, P. (eds.), *Anticarcinogenesis and Radiation Protection*. New York, Plenum, 1988.
2. Puck, T. T., Morkovin, P. I., Marcus, S. J., et al. Action of x rays on mammalian cells II. Survival curves of cells from normal human tissue. *J. Exp. Med.* 1957; **106**:485.
3. Lea, D. E. *Actions of Radiation on Living Cells*, 2nd edition. New York, Cambridge University Press, 1955.
4. Haynes, R. H. The interpretation of microbial inactivation and recovery phenomena. *Radiat. Res. Suppl.* 1966; **6**:1.
5. Chapman, J. D. Biophysical models of mammalian cell inactivation by radiation, in Meyn, R. E., and Withers, H. R. *Radiation Biology in Cancer Research*. New York, Raven Press, 1980, p. 21.
6. Alpen, E. L. *Radiation Biophysics*, 2nd edition. San Diego, CA, Academic Press, 1998.
7. Johns, H. L., and Cunningham, J. R. *The Physics of Radiology*, 4th edition. Springfield, IL, Charles C Thomas Publishers, 1983, p. 680.
8. Sinclair, W. K., Morton, R. A. X-ray sensitivity during the cell generation cycle of cultured Chinese hamster cells. *Radiat. Res.* 1966; **29**:450-474.
9. Strandqvist, M. Studien ber üdie kumulative Wirkung der Röntgenstrahlen bei Fraktionierung. *Acta Radiol. Suppl.* (Stockh.) 1944:55.
10. Ellis, F. Dose, time and fractionation: A clinical hypothesis. *Clin. Radiol.* 1969; **20**:1.
11. Hall, E. J. *Radiobiology for the Radiologist*, 3rd edition Philadelphia, J. B. Lippincott, 1988.
12. Barth, R. F., and Soloway, A. H. Boron neutron capture therapy of cancer. *Cancer Res.* 1990; **50**:1061-1070.
13. Vriesendorp, H. M., and vanBekkum, D. W. Role of total body irradiation in conditioning for bone marrow transplantation. *Haematol. Bluttransfus.* 1981; **25**:349-364.
14. Lusbaugh, C. C. Reflections on Some Recent Progress in Human Radiobiology, in Augenstein, L. G., Mason, R., and Zelle, M. (eds.), *Advances in Radiation Biology*. New York, Academic Press, 1969, pp. 277-314.
15. Looney, W. B., Trefil, J. S., Hopkins, H. A., et al. Solid tumor models for the assessment of different treatment modalities: Therapeutic strategy for sequential chemotherapy with radiotherapy. *Proc. Natl. Acad. Sci. USA* 1977; **74**:1983-1987.
16. Fowler, J. F., Denekamp, J., Page, A. L., et al. Fractionation with x-rays and neutrons in mice: Response of skin and C3H mammary tumors. *Br. J. Radiol.* 1972; **45**:237-249.
17. Travis, E. L., Vojnovic, B., Davies, E. E., et al. A plethysmographic method for measuring function in locally irradiated mouse lung. *Br. J. Radiol.* 1979; **52**:67-74.
18. Van der Kogel, A. J. Mechanisms of Late Radiation Injury in the Spinal Cord, in Meyn, R. E., and Withers, H. R. (eds.), *Radiation Biology in Cancer Research*, New York, Raven Press, 1980, pp. 461-470.

19. Morgan, K. Z., and Turner, J. E. *Principles of Radiation Protection*. New York, John Wiley & Sons, 1967, p. 422.
20. Hendee, W. R. Radiation Carcinogenesis, in Hendee, W. R., and Edwards, F. M. (eds.), *Health Effects of Exposure to Low-Level Ionizing Radiation*. Philadelphia, Institute of Physics Publishing, 1996, pp. 103–111.
21. BEIR V, National Academy of Sciences, Committee on the Biological Effects of Ionizing Radiations. *Health Effects of Exposure to Low Levels of Ionizing Radiation*, Report No. 5. Washington, D.C., National Academy Press, 1980.
22. Selby PB, Selby PR. Gamma-ray-induced dominant mutations that cause skeletal abnormalities in mice. I. Plan, summary of results and discussion. *Mutat. Res.* 1977; **43**:357–375.
23. Selby, P. B. Genetic Effects of Low-Level Irradiation, in Fullerton, G. D., Kopp, D. T., and Webster, E. W. (eds.), *Biological Risks of Medical Irradiations*. New York, American Institute of Physics, 1980, pp. 1–20.
24. Muller, H. J. The effects of x-radiation on genes and chromosomes. *Science* 1928; **67**:82.
25. Sankaranarayanan, S. Radiation Mutagenesis in Animals and Humans and the Estimation of Genetic Risk, in Hendee, W. R., and Edwards, F. M. (eds.), *Health Effects of Exposure to Low-Level Ionizing Radiation*. Philadelphia, Institute of Physics Publishing, 1996, pp. 113–167.
26. Russell, W. L. X ray induced mutations in mice. *Cold Spring Harbor Symp. Quant. Biol.* 1951; **16**:327–336.
27. Russell, W. L., Russell, L. B., and Kelly, E. M. Radiation dose rate and mutation frequency. *Science* 1958; **128**:1546–1550.
28. Russell, W. L., and Kelly, E. M. Mutation frequencies in male mice and the estimation of genetic hazards of radiation in men. *Proc. Natl. Acad. Sci. USA* 1982; **79**:542–544.
29. Neel, J. V., and Schull, W. J. (eds.), *The Children of Atomic Bomb Survivors: A Genetic Study*. Washington, D.C., U.S. National Academy of Science, 1991.

HUMAN RADIOBIOLOGY

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■ OBJECTIVES

After completing this chapter, the reader should be able to:

- Define the term “stochastic effects” and name the three effects that have been identified as stochastic radiation effects.
- Name the subcategories of the acute radiation syndrome (ARS).
- Define the term “collective dose equivalent.”
- Define the term “effective dose.”
- List the sources of natural and manmade background radiation and name the largest contributors to each.
- List 10 examples of human populations that are or have been exposed to unusual levels of radiation.
- Discuss current dose-effect models and the factors that influence them.
- Give an estimate of the risk per unit effective dose of fatal cancer due to a single exposure to ionizing radiation.

■ STOCHASTIC EFFECTS OF RADIATION

It is common knowledge that the incidence of leukemia among atomic bomb survivors in Hiroshima and Nagasaki began to increase a few years after the explosions. This increase has been associated with exposure to ionizing radiation produced by the blasts. Leukemia is one example of a long-term consequence of radiation exposure.

Radiation effects that appear several years or more after individuals are irradiated have several characteristics in common:

- The probability of occurrence of the effect (i.e., the fraction of persons in a population who exhibit the effect) increases with dose.
- The severity of the effect in a single individual is unrelated to the magnitude of the dose.
- There is no threshold below which it can be said with certainty that the effect will not occur.

Clearly, these effects are a result of radiation activation of mechanisms within the body that produce disease. A stochastic process (disorder or radiation effect) is one in which the probability of occurrence, but not the severity of the disorder, is related to the dose of radiation. The fact that severity is unrelated to dose suggests that radiation triggers a disease process whose ultimate course is determined by other factors in the individual.

One of the difficulties in providing precise estimates of the role of an agent such as radiation in producing stochastic effects is that these effects also occur in the absence of irradiation. That is, there is a “natural incidence” of the effects due to causes other than the agent under consideration. The three stochastic effects of radiation that have been identified to date are:

- Carcinogenesis, the induction of cancer
- Teratogenesis, the induction of birth defects by irradiation of the fetus
- Mutagenesis, the induction of genetic disorders in future generations by irradiation of germ cells (sperm and ova) prior to conception

■ NONSTOCHASTIC EFFECTS OF RADIATION

Most of the “immediate” effects of radiation—that is, those that occur within several weeks of radiation exposure—are associated with dose levels far above those delivered in modern diagnostic radiology. One set of such effects is known

In one case of fluoroscopic overexposure of the patient’s skin, the patient’s arms were at her side during a cardiac catheter ablation procedure in which the x-ray tube was repeatedly maneuvered such that the right humerus was plainly visible in the center of the beam. In addition, a spacer had been removed from the x-ray tube housing, allowing the tube to be much closer than usual to the x-ray tube. It has been estimated that the total dose to the arm exceeded 25 Gy. *Source:* Wolff D, Heinrich KW. Strahlenschaden der haut nach herzkatheterdiagnostik und -therapie 2 kasuistiken. *Hautnah Derm.* 1993; 5:450–452.

As of 2001, 71 cases of radiation injury to the skin are known, with most associated with cardiac catheterization. Even assuming that many have gone unreported, the true number is undoubtedly a very small fraction of the approximately 30,000 interventional procedures performed yearly in the United States.

colloquially as “radiation sickness” or, more formally, as acute radiation syndrome (ARS).

The acute radiation syndrome is divided into three subcategories: gastrointestinal (GI), hematopoietic, and cerebrovascular syndromes. These subcategories reflect both the relative sensitivities of various organ systems to radiation and the time required to produce effects on the overall health of the organism.

Hematopoietic Syndrome

The stem cells of the hematopoietic system, precursors of mature blood cells that are found principally in the bone marrow, can be inactivated in significant numbers at doses of a few gray. Loss of the precursor population may not pose a significant threat to the individual until it is time for the precursor cells to reach maturity. The body may then lose the ability to combat infection. Thus, a latency period may occur for a week to a month, during which attempts may be made to reestablish the bone marrow population through transplants or other techniques.

Gastrointestinal Syndrome

The cells associated with the GI tract, particularly epithelial cells that line the intestinal surfaces, are particularly susceptible to radiation damage. Following an exposure of several gray or more, diarrhea, hemorrhage, electrolyte imbalance, dehydration, etc., occur as a consequence of cell damage.

Cerebrovascular Syndrome

At doses above 50 gray, damage to the relatively radiation insensitive neurologic and cardiovascular compartments is severe enough to produce an almost immediate life-threatening syndrome. Death in this syndrome is brought about by destruction of blood vessels in the brain, fluid buildup, and neuronal damage. Death may occur in a few days or, at higher doses (greater than 100 Gy), within hours.¹ Because of the involvement of brain and nervous system tissues, the cerebrovascular syndrome (CVS) is sometimes referred to as central nervous system syndrome. The hematopoietic and GI systems are also severely damaged at this dose level but would not contribute to the cause of death in such a short time.

Fertility and sterility are also affected at high dose levels. Temporary sterility may occur in the male for testis irradiation above 0.15 Gy. Permanent sterility results from doses exceeding 3.5 Sv.² In the female, temporary sterility has been observed for ovarian doses above 0.65 Gy, and permanent sterility for doses above 2.5 Gy.³ These thresholds are for single instantaneous exposures. Doses required to produce temporary or permanent sterility are increased by orders of magnitude if delivered either in fractions or at lower dose rates.

One nonstochastic effect that occurs within weeks of radiation exposure that may be encountered in diagnostic radiology is the skin effect. In 1994, the FDA issued an “advisory,” a document intended to warn health care practitioners that certain fluoroscopic procedures (cardiac ablation, coronary angioplasty, percutaneous transluminal angioplasty, and a few others) had been found, in rare instances, to produce skin effects.⁴ In some cases, this was reddening of the skin that eventually ceased. However, in some cases, the skin became necrotic and multiple skin grafts were required. All were associated with the use of “high level control” mode, a mode of fluoroscopy in which a dose of as much as 10 times higher than normal is used to decrease noise so that moving catheters may be seen without exceeding the patient’s level of tolerance for contrast material. In most cases the total fluoroscopy “beam on” time was excessive, due to complications or the need for multiple procedures. One factor that has caused this fairly recent phenomenon is the ability of modern x-ray tubes to tolerate greater heat loads so that procedures can go on much longer than a decade ago. Table 27-1 gives the doses required to obtain various skin reactions. At typical

Because of the radionuclides contained in our bodies, our cells are irradiated by 30 million radioactive decays per hour. An ionizing interaction occurs in every cell in our body at least once a year.

TABLE 27-1 Radiogenic Skin Effects^a

<i>Skin Effect</i>	<i>Approximate Single-Dose Threshold (Gy)</i>	<i>Approximate Time to Onset</i>	<i>Peak</i>
Transient erythema	2	a few hours	24 hr
Temporary epilation	3	3 weeks	NA
Main erythema	6	10 days	—
Dry desquamation	10	4 weeks	5 weeks
Moist desquamation	15	4 weeks	5 weeks
Late erythema	15	6–10 weeks	NA
Dermal necrosis	18	>10 weeks	NA
Ulceration	20	>6 weeks	NA

^aData from Wagner, L. K., Eifel, P. J., and Geise, R. A. Potential biological effects following high x-ray dose interventional procedures. *J. Vascular Interventional Radiol.* 1994; 5:71–84.

fluoroscopic dose rates (~ 0.02 Gy/min at the entrance surface), erythema would probably not occur unless 1.7 hours of fluoroscopy beam-on time were exceeded. However, at high-level dose rate (~ 0.20 Gy/min), only 10 minutes is required.

Radiogenic skin injury from this source is especially challenging in terms of medical management. Some patient's reports of skin problems were initially treated as nonspecific dermatitis or some other nonradiogenic disorder because the injury was so far removed in time from the fluoroscopic procedure and dermatologists were unfamiliar with fluoroscopically induced skin damage. It is assumed that many cases of reversible skin changes have gone unreported.

Because of the potential for serious harm, the FDA recommends that patient dose be monitored if a procedure has the potential for delivering more than 1 Gy. Following the procedure, some indication of the specific areas of skin and the doses estimated for those areas is to be entered into the patient's chart. Some indication is also to be included regarding the types of skin effects that might be seen.

Although the majority of procedures for which skin effects have been demonstrated have been cardiac, neurointerventional procedures also involve patient doses that would put them within the criteria for monitoring and including notes in the patient's record. A review of over 500 neurointerventional procedures at a major medical center where the x-ray equipment was equipped with an automated dosimetry system demonstrated that 6% of embolization procedures and 1% of cerebral angiograms resulted in skin doses exceeding the erythema threshold of 6 Gy.⁵

■ DOSIMETRY IN INDIVIDUALS AND POPULATIONS

The dosimetry of ionizing radiation as it concerns the individual patient or irradiated specimen has been discussed elsewhere (Chapter 7). Estimation of radiation risk to individuals within populations requires some additional terminology. At the core of this terminology is the assumption that the effects of radiation at low doses are stochastic (probabilistic for the occurrence of effect, with the severity of effect not related to dose). As far as the individual is concerned, an increase in dose increases the probability of occurrence of an adverse effect. In terms of dose to a population, when more individuals receive a given dose, a greater number exhibit an adverse effect. The collective dose equivalent (CDE), defined as the sum of the dose equivalents for individuals in a population, has units of person-sievert (or person-rem). The average dose equivalent (ADE), which is the CDE divided by the total number of individuals in a population, has units of sievert (or rem).

The modifiers “committed” and “effective” are sometimes added to CDE and ADE. Committed dose refers to the dose delivered by radioactive materials deposited internally within the body. Such materials deliver radiation over a period of time determined by the biological and physical half-lives of the radionuclide. Thus, by

summing (integrating) the dose delivered during a series of time intervals, the total commitment of dose over the life span of an individual may be determined.

To describe the overall harm to an organism, one must consider the varying degrees of damage caused by the irradiation of different organs.⁶ Harm, in this context, refers to the risk of cancer and hereditary effects. As an index of overall harm, the effective dose (ED) concept uses weighting factors for each organ. These weighting factors, which reflect the radiosensitivity of the organ and its importance to the organism as a whole, are applied to doses delivered to each organ. For example, the weighting factor for radiation delivered to the breast (0.15) is higher than the weighting factor for a dose delivered to the thyroid (0.03). This concept was discussed extensively in Chapter 6.

Immediate effects of radiation exposure are also referred to as *early effects*; delayed effects are frequently termed *late effects* of radiation exposure.

■ BACKGROUND RADIATION

All of the earth's inhabitants are exposed to background radiation. Some of this radiation is man-made, such as radiation used in medical applications, and some is "natural." Natural sources include cosmic rays, terrestrial, and internal. Man-made radiation includes medical x-rays, medical nuclear procedures, consumer products, industrial sources, and some miscellaneous sources of radiation. A typical U.S. resident receives a total of 3.6 mSv (360 mrem) in one year from all sources of background radiation.⁷ The actual background encountered by any one individual varies significantly, depending upon where he or she lives, the food that is consumed, the radon levels in the home, and so on.⁸

Natural Background

Cosmic rays have always bombarded the earth. A typical U.S. resident receives 0.29 mSv (29 mrem) from cosmic rays. The earth's atmosphere provides some shielding from cosmic rays. This shielding is reduced at higher elevations, and the cosmic ray dose is increased. Inhabitants of Denver, Colorado, at an elevation of 1600 meters (1 mile), receive 0.50 mSv/yr (50 mrem/yr) from cosmic rays, while those in Leadville, Colorado, at an elevation of 3200 meters (2 miles), receive 1.25 mSv (125 mrem). The average value of 0.29 mSv reflects the distribution of the U.S. population living at different elevations.

Terrestrial background originates from radioisotopes that are found everywhere in our surroundings. All elements found in nature have radioactive isotopes, many of which are also present in the environment. The exact composition of soil influences the local terrestrial background, because the minerals present determine which elements are most abundant. Terrestrial background sources are categorized as "primordial" if their half-lives are the same order of magnitude as the presumed lifetime of the earth (4.5×10^9 years). That is, these sources were present when the earth was formed and there is no way to replenish them in nature. Two isotopes of uranium, ^{238}U and ^{235}U , and one of thorium ^{232}Th , give rise to three different decay series. In each of these series, the radioactive nuclide decays to another stable isotope of bismuth or lead. Seventeen other nuclides are primordial, but are not part of a decay series. Of these nonseries radionuclides, ^{40}K and ^{87}Rb make the greatest contribution to background dose.

Internal background is the dose imposed by the isotopes contained in our bodies. A small percentage of the potassium in the human body is ^{40}K . This radioactive nuclide emits both locally absorbed beta radiation and more penetrating gamma radiation. Similarly, ^{14}C , which comprises a small percentage of the carbon atoms found in organic molecules throughout our bodies, contributes to a total dose of 0.39 mSv/yr (39 mrem/yr) to the average U.S. resident from internal background.

Most texts consider radon exposure to be a part of natural background, although it could be argued that there is a component of human activity involved in exposure to radon. As part of the ^{238}U decay series, radon (specifically, ^{222}Rn , referred to hereafter as simply radon) is significant because it is an alpha emitter that exists as an inert gas.

The ICRP, NCRP, and BEIR Committee are voluntary efforts, and their recommendations are purely advisory. However, in the United States, NCRP recommendations are frequently codified into radiation regulations by federal and state agencies, including the Nuclear Regulatory Commission and the Environmental Protection Agency.

It is estimated that, anywhere on earth, a million-ton sample of the earth's crust would contain four tons of uranium (also, for comparison, 70 tons of copper and 10 lbs of gold).

Stochastic radiation effects are those effects for which the probability of occurrence of the effect (as opposed to the severity of the effect) increases with dose. Although at small doses the probability of occurrence may be very small, it is presumed to never reach zero. That is, it is assumed that there is no threshold dose. Stochastic effects are sometimes called *probabilistic effects*.

Since it is inert, radon generated by decay of ^{226}Ra at some depth in the soil does not bind chemically with other elements. Instead, it percolates up to the surface to escape into the atmosphere. Being heavier than most constituents of the atmosphere, it tends to remain at lower elevations. Although minable deposits of uranium ore are primarily associated with granite rock formations, uranium is found everywhere in the earth's crust. Ninety-nine percent of the uranium found in nature is ^{238}U . Thus, the air we breathe anywhere on earth contains some amount of radon.

Radon itself is not particularly hazardous, when inhaled, because it does not react and in most cases is simply exhaled. A more significant concern is that two of the decay products of radon (^{218}Po and ^{214}Po), delivered to the air by decaying radon, are not inert. These products adhere to dust particles in the air, which may then be inhaled into the lung. The two alpha emitting isotopes, ^{218}Po and ^{214}Po , account for 85% of the dose to the lung from "radon;" two other beta and gamma emitting isotopes in the series, ^{214}Pb and ^{214}Bi , account for most of the remaining 15%. A typical U.S. resident receives an annual dose equivalent of 2.0 mSv (200 mrem) from radon and its decay products.⁷

The exact value of dose equivalent received by an individual is most heavily influenced by the location where the person spends his or her time—buildings. With a combination of an unusually strong source (e.g., an unfinished crawl space below a house in a region with a high concentration of uranium in the soil) and air flow (e.g., radon buildup due to the lack of air exchange in an energy-efficient house, or radon buildup because of a "chimney effect" in a drafty house), some homes can have radon levels that are significantly elevated. Although the risk to any one individual is very small, many people (i.e., the entire population) are exposed to some level of radon. By applying the linear, no-threshold model of radiation-induced cancer, one can estimate that a significant number of lung cancers may be caused each year by long-term exposure to high radon levels. Therefore, the U.S. Environmental Protection Agency recommends that radon concentrations be measured in buildings and that remedial action be taken if the radon concentrations exceed 4 pCi/l of air. This level of radon has been estimated to yield (a) a lung dose in a typical adult of 0.05 mGy (5 mrad) and (b) an increase in the risk of lung cancer of 2% for each year of exposure.

Man-Made Background

Various human activities add to the annual radiation background. Chief among these is diagnostic radiology (0.39 mSv, 39 mrem), and second is nuclear medicine (0.14 mSv, 14 mrem).⁷ Radiation received by patients in radiation therapy is not counted in man-made background, because the intention is to track radiation doses that are associated only with stochastic effects. The large doses received in radiation therapy are associated with both deterministic and stochastic effects. This omission has little impact on the average level of man-made background, because the number of persons who receive radiation treatment is a very small fraction of the U.S. population.

Of course, in any given year an individual may or may not undergo a medical diagnostic study. The actual dose received by an individual from a medical procedure can be estimated from clinical information such as the measured output of the x-ray tube, patient thickness, technique factors, and so on. The concern that is addressed by calculating the population average of 0.39 mSv per person per year is the collective equivalent dose received by the entire population, since it is the collective dose that is responsible for population bioeffects such as increases in somatic and genetic mutation rates.

Various consumer products emit small amounts of radiation.⁹ Some examples include exit signs that contain ^3H , a low-energy beta emitter, and smoke detectors that contain ^{241}Am , an alpha emitter. Luminous dials on watches, clocks, and instruments contained ^{226}Ra at one time but, since 1968, have been completely replaced by

^3H and ^{147}Pm , both of which are low-energy beta emitters. The low-energy beta particles emitted by these substances are absorbed in the instrument components and provide negligible amounts of radiation dose to their owners. Increasingly, liquid crystal displays and light-emitting diodes are replacing the use of radioactive materials in luminous displays. Collectively, consumer products are estimated to contribute approximately 0.1 Sv (10 mrem) to the yearly dose from man-made background radiation.

Other contributions to man-made radiation include (a) aspects of the nuclear fuel cycle, ranging from mining to transportation and to waste storage, (b) the occupational exposure that a relatively small fraction of the population receives, and (c) fallout from nuclear weapons tests. These components are thought to account for approximately 0.02 mSv (2 mrem) per year.

The United States and other countries detonated over 500 nuclear weapons above ground during the time period 1945–1963. This activity had a measurable effect upon natural background radiation levels. In a nuclear explosion, new elements are created through the processes of nuclear fission and fusion. These elements are injected into the upper atmosphere by the force of the blast, where they drift over large distances before eventually returning to earth. The United States and the former Soviet Union ceased above-ground testing in 1963. A few countries conducted above-ground tests after that time, but by 1980 all above-ground testing had stopped. Individuals who were alive during the time period 1945–1980 received radiation doses from inhalation and ingestion of some atoms of radioactive material as a result of these tests. Calculation of this dose involves complicated functions of rate of uptake, rate of decay of the nuclides in individuals, rate of insertion of new isotopes into the environment, and other factors. The United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) has estimated that an individual who was born before 1945 has received a total of 4.5 mSv (450 mrem) until now because of nuclear testing. The current dose rate is estimated to be 0.01 mSv/yr (1 mrem/yr).¹⁰

HUMAN POPULATIONS THAT HAVE BEEN EXPOSED TO UNUSUAL LEVELS OF RADIATION

Atom Bomb Survivors

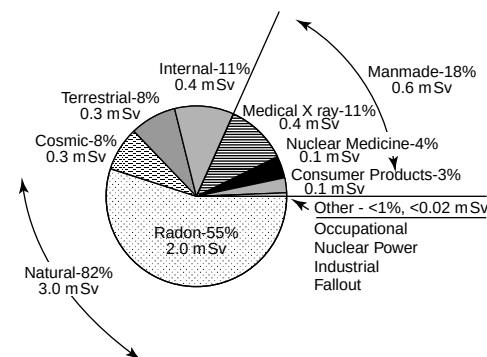
The atomic bomb blasts at Hiroshima and Nagasaki left a population of nearly 280,000 persons who survived the immediate effects of the blast but were subjected to the long-term consequences of exposure to ionizing radiation.^{11–17} This population represents the highest CDE for any large population to date and is the most extensively studied of all groups of exposed individuals. Of the survivors, dosimetry estimates are available for approximately 80,000. Of that group, 42,000 are estimated to have received greater than 0.005 Gy (0.5 rad), with the rest serving as controls. Cancer incidence and the effects of radiation exposure *in utero* have been documented in this population. Much of the risk estimate data discussed elsewhere in this chapter is based upon studies of atom bomb survivors.

Early Radiologists, Nurses, and Technologists

Radiologists in the early part of the twentieth century often worked with x-ray equipment that lacked safety features such as lead shielding and collimation.^{18–20} Safety standards did not become commonplace until after World War II. Although it is difficult to reconstruct dosimetry for individuals, this group of radiation workers has been studied to demonstrate the need for current safety standards. The incidence of leukemia, for example, has been found to be elevated for radiologists who experienced a significant part of their careers prior to World War II.²¹

Nonstochastic radiation effects are those effects that exhibit a dose threshold. After the threshold is exceeded, the severity of the effect increases as the dose increases. Nonstochastic effects are sometimes called *deterministic effects*.

Radon was first identified as a significant source of background exposure when a worker reported to the Limerick Nuclear Power Plant in Pennsylvania for his first day of work in 1984. Radiation detectors were located at all exits from the plant to detect any contamination resident upon workers as they left. This worker set off radiation detectors when he entered the plant. Subsequent investigation revealed an unusually high concentration of uranium near the foundation of his home.



MARGIN FIGURE 27-1

A typical U.S. citizen's yearly background radiation equivalent dose, broken down into various natural and manmade sources. (From National Council on Radiation Protection and Measurements (NCRP) Report No. 93, *Ionizing Radiation Exposure of the Population of the United States*, NCRP, Bethesda, MD, 1987.)

Uranium and Other Miners

It has been known for many years that the “mountain sickness” reported in miners as early as the fifteenth century was probably lung cancer.^{22–25} The excess rates of cancer were probably initiated by inhalation of particulates with attached radioactive atoms such as the decay products of radon. Current estimates of radon hazards in the home have arisen in part from studies of uranium miners.

Radium Dial Painters

About 1% of radium dial painters who worked in the industry between 1913 and 1925 died of bone and sinus cancer. This is 33% higher than the rate found in control populations. In 1925, the industry passed a rule stating that the paint brush containing the radium paint shall never be placed in the mouth. Cancer rates for workers who started work after 1925 have been found to be equal to those of control groups.

In the period from 1915 to 1930, approximately 1500 workers, mostly women, were employed to paint watch dials with a luminous compound containing the α -emitter radium (^{226}Ra).^{26–30} This activity was conducted before modern safety standards for handling radioactive materials were commonplace. Many of the workers ingested or absorbed radium compounds through work habits such as using the mouth to bring the tip of paint brushes to a point. Excess numbers of carcinomas of the paranasal sinuses and mastoid air cells have been observed in this population of workers. These cancers are thought to have arisen from irradiation of the tissue surrounding air cavities containing radon gas (^{222}Rn) and its decay products, and from direct irradiation of underlying bone by the radium, a chemical analogue of calcium that is taken up by bone and incorporated into the skeleton. Of the radium dial painters, approximately 35 cases of osteogenic sarcomas and sinus carcinomas have been identified over a follow-up period exceeding 50 years.

Radiation Therapy Patients

It is difficult to determine if cancer patients who have been treated with radiation are at increased risk for cancers occurring specifically as a result of the treatment. Studies have been attempted, but are hampered by the existence of proven cancers in those populations and the fact that second cancers are more prevalent inherently in patients who have had an earlier cancer. However, a number of benign disorders have been treated in the past with radiation at dose levels of several gray and above. These include tinea capitis (ringworm), thymic enlargement, ankylosing spondylitis, post-partum mastitis, and menorrhagia. Positive correlations between radiation exposure and several types of subsequent cancer, particularly skin cancer, have been shown in these populations.^{31–36}

Diagnostic Radiology Patients

Radiation-induced changes in the vascular endothelium are thought to be the underlying cause of most nonstochastic radiation effects.

In the past, some diagnostic studies were performed on patients with techniques that yielded exposures that were orders of magnitude greater than those for similar procedures today.^{37–43} For example, fluoroscopy was used from 1930 to 1954 to monitor pneumothoraces induced as part of the treatment for tuberculosis. Radiation doses of up to 6 Gy from this procedure have been associated with an increased incidence of breast cancer.

Most population studies of patients that have received doses on the order of those used currently have failed to provide statistically significant evidence of harmful effects. For example, a study of more than 10,000 ^{131}I thyroid examinations performed in Sweden in the 1950s failed to show an increase subsequent in thyroid cancer.³⁹

Some marginal associations between diagnostic radiation exposure and adverse health effects remain. Retrospective studies of leukemia patients have examined the evidence of a link between adult-onset myelogenous leukemia and x-ray examinations. Some correlations have been found, particularly for individuals who experienced multiple radiographic examinations. There is also some evidence that *in utero* exposure increases the risk of childhood leukemia. A study of prenatal x-ray

exposure in asymptomatic twin pregnancies (a technique that was used until the early 1970s to determine fetal position, resulting in a fetal dose of a few rem) demonstrated an increased incidence of childhood leukemia for the exposed twins by 1 chance in 2000. However, questions remain regarding these studies related to the possibility of statistical bias, inadequate dosimetry, and statistical power.

Nuclear Weapons Tests

In addition to the Japanese A-bomb survivors, a number of individuals were exposed to radiation as a result of the detonation of nuclear weapons.^{44–46} These individuals include military personnel who conducted maneuvers in the vicinity of detonations, and residents of areas “downwind” of test sites. Epidemiologic studies among these groups are difficult because of inexact dosimetry. Also, organ doses for most of these are estimated to be below 50 mGy (5 rem). Thus it is not surprising that there is little statistically significant correlation between radiation exposure and adverse health effects in most of these groups.

Adverse health effects have been detected in individuals whose organ doses may have been higher than 50 mGy. These individuals include some residents near the Nevada test site, and Marshall Islanders in the Pacific who were accidentally exposed to fallout from a thermonuclear test explosion in 1954. Ingestion of ¹³¹I and external exposure have been associated with an increased rate of nodule formation and cancer in the thyroids of the Marshall Islanders.

Residents of Regions with High Natural Background

The presence of naturally occurring radioactive materials in the soil has resulted in increased natural background radiation by inhabitants in many parts of the world.^{47–52} Some of the most extensive epidemiologic studies of native populations have examined the effects of monazite, a mineral containing thorium, uranium, and radium. The global average natural background is approximately 1 mGy/yr, excluding exposure from radon and radon progeny in the lungs. Because of monazite, the natural background for the 12,000 inhabitants of Guarapari, Brazil is 6.4 mGy/yr, for the 70,000 individuals in the Kerala Coast of India it is 3.8 mGy/yr, and for the Guangdong Province of the People's Republic of China it is 3.5 mGy/yr. Studies of these populations have repeatedly shown an increased frequency of chromosomal aberrations. However, a heightened incidence of genetic disorders has not been unequivocally demonstrated. In the United States, the relatively smaller range of natural background radiation, coupled with the mobility of the population, has created difficulties for studies of the effects of background radiation upon regional morbidity and mortality. Studies have shown, as expected, no correlation.

Air and Space Travel: Encountering Cosmic Rays and Solar Flares

Cosmic rays are composed of various types of radiation that are emitted into space throughout the universe by supernovae, quasars, nebulae, and so on. The earth's atmosphere, along with the magnetic fields that surround the planet, act to shield the surface of the earth from most of the cosmic rays. On the surface of the earth, cosmic rays are nearly 100% protons. At sea level, the equivalent annual dose from cosmic rays is approximately 0.3 mSv (30 mrem). This figure approximately doubles for every mile above sea level. The dose also increases with latitude, because the earth's magnetic field “steers” charged particles toward the poles. A flight in a typical commercial airliner results in an equivalent dose rate of approximately 0.005 to 0.01 mSv/hr (0.5 to 1 mrem/hr), depending upon altitude and latitude. Outside of the earth's magnetic field, cosmic rays are 80% protons, 12% alpha particles, and 8% ions of practically all of the naturally occurring elements, with the greatest contribution

Monazite sand formations in Guarapari, Brazil were expected to produce elevated natural background levels for inhabitants of that region due to a higher concentration of thorium, uranium, and radium than is found in most soils. The amount of this radiation actually received would depend upon how much time people spent outdoors, what kind of food they ate, and so on. Dr. Thomas L. Cullen, a physicist and Jesuit priest at the Catholic University of Rio de Janeiro, wanted to measure background radiation levels in this population. He needed to convince a large sample of the rural population to wear radiation dosimeters for three months as they went about their normal lives. Reasoning that the majority of inhabitants were devout Catholics, he had hollow medals of the Virgin Mary filled with thermoluminescent dosimetry powder. After they were blessed by Dr. Cullen, he was able to obtain the cooperation of 115 people. The medals were opened three months later and the powder was returned to Dr. Cullen. The subjects were allowed to keep the medals. This statistical sample was still somewhat biased as most non-Catholics in the region refused to wear the medals.

Source: John Cameron, *Health Physics Society Newsletter*, vol. xxix, No. 3, p. 11, March 2001.

The publication of *Silent Spring* by Rachel Carson in 1962 reinforced the public's awareness about the potential health consequences of environmental agents such as fallout from nuclear weapon blasts (atomic tests) and pesticides.

Space travel is associated with a risk of dangerously high radiation exposure. The astronauts who traveled to the moon in the 1970s carried radioprotective drugs. These drugs, while able to reduce the effects of radiation by almost a factor of three (so that three times as much radiation would be required to produce the effect once the drug had been taken), were also quite toxic. Among other side effects, they reduce the blood pressure to dangerously low levels. If a large solar flare had occurred during a lunar mission, they would have taken the drug, because the amount of radiation they would have received could conceivably have been lethal. Source: Hall, E. J., *Radiobiology for the Radiologist*, Lippincott, New York, pp. 204–205.



MARGIN FIGURE 27-2

The Space Shuttle usually flies in such a way that radiation dose is minimized, by avoiding the poles and the South Atlantic region and keeping the altitude below 500 kilometers (~300 miles). During a typical mission of the Space Shuttle, the astronauts receive approximately 1 mSv (100 mrem) per day. This plot of the orbit of the Space Shuttle Discovery on March 9, 2001 shows that the South Atlantic Anomaly (SAA), a region in which the Van Allen radiation belts exist at lower altitude, has been avoided.

in dose coming from iron ions. Such “heavy” ions ($Z_{\text{iron}} = 56$) have extremely high LET.

Sunspots are dark spots that occasionally appear on the surface of the sun. They indicate regions of increased electromagnetic activity and are sometimes responsible for ejecting particulate radiation into space. This ejected radiation is called a solar flare. Solar radiation normally constitutes a small fraction of the dose from cosmic radiation on earth. However, the solar contribution to the cosmic ray background increases during periods of high sunspot activity. A person flying 10 hours aboard a commercial aircraft during a period of normal sunspot activity will receive a radiation dose equivalent that is approximately equivalent to that received from a single chest x-ray examination. During a solar flare, this cosmic-ray dose can be 10 to as much as 100 times greater. Still, this increase in radiation exposure carries an immeasurably small health risk. The Atomic Bomb survivor studies revealed no excess cancers in subjects who received less than about 35 rem (350 millisievert).

In space, the same rules for dose from cosmic rays apply as for earthbound individuals: Higher elevation and higher latitude (more northerly) means higher dose from cosmic rays and solar flares. In addition, astronauts may encounter the Van Allen radiation belts. These belts are clouds of charged particles (mostly electrons and protons) that collect in specific, stable patterns around the earth. They are not perfect spherical shells because of the motion of the earth, the effect of the earth's magnetic field, and the effect of the solar “wind,” streams of charged particles that emanate from the sun. The Van Allen belt comes closest to the earth in a region over the South Atlantic. There are actually several “belts”; but the main one, in terms of radiation dose, begins at a height of approximately 18,000 km and extends to a height of approximately 80,000 km.

The Space Shuttle usually flies to minimize radiation dose by avoiding the poles and the South Atlantic region and by keeping the altitude below 500 kilometers (~300 miles). During a typical mission of the Space Shuttle, the astronauts receive approximately 1 mSv (100 mrem) per day.

Nuclear Accidents

A few persons have received large doses of radiation in the course of accidents involving critical masses of fissionable materials or accidentally discarded radioactive sources.^{53–55} Two nuclear power reactor accidents in recent years that have generated considerable publicity actually produced very different outcomes. The Three-Mile Island (TMI) reactor accident in March 1979, although it had the potential for severe impact, actually resulted in negligible exposure of employees and persons living near the reactor. The average dose to the population of 2 million persons living within 50 miles of TMI was 20 μSv (2 mrem). No detectable increase in cancer deaths is expected for this population, and no adverse health effects, with the exception of anxiety and possibly stress disorders, have been identified in this population. The Chernobyl nuclear reactor accident in April 1986, on the other hand, resulted in whole-body doses exceeding 1 Gy (100 rad) to over 200 workers. Thirty workers who received over 4 Gy (400 rad) subsequently died as a result of a combination of radiation exposure and fire-related injuries. There was a significant increase above background radiation to more than 75 million Soviet residents, with the highest exposure occurring in a population of 272,800 living within 200 miles of the reactor. More than 100 million curies of radioactivity was released from the reactor. Human exposures were caused primarily by the release of approximately 17 million curies of ^{131}I and 2 million curies of ^{137}Cs . For comparison, 15 curies of ^{131}I were released in the TMI accident. The radioactive release from Chernobyl resulted in a total collective dose of 55,000 person-sievert (5.5 million person-rem) to the population of 272,800 persons, yielding an average dose of 0.2 Sv (20 rem). In some individuals, thyroid doses caused by the ingestion of milk could have exceeded several hundred rem. The long-term health consequences in the population are not yet fully known.

DOSE-EFFECT MODELS

Many factors influence the effects of radiation upon humans. It is tempting to speculate that if a complete list of all possible “lifestyle” factors such as socioeconomic status, diet, exposure to chemicals at work and in the home, and detailed biochemical and genetic information were available for an individual, one could then compute with reasonable certainty the probability of occurrence of a particular radiation-induced stochastic health effect for that individual. Even if complete knowledge were available, however, the computed result would still be a probability of effect because the nature of radiation interactions and their effect on tissues are probabilistic in nature. Furthermore, to make this calculation one would need to know the effects of each of the lifestyle and biochemical factors and whether they operate independently or in concert (synergistically) with other factors in the list.

At the present time, knowledge concerning the role of many of the factors listed above is incomplete. Undoubtedly, major advances will occur in the near future in the understanding of fundamental biochemical mechanisms such as the role of tumor promoters and inhibitors, co-carcinogenic interactions, and the role of genetics. For the present, estimates of the radiation risk to individuals are based primarily upon epidemiologic evidence. From this evidence, efforts have been directed to determine the role of several factors, principally dose, dose rate (including dose-modifying factors described in greater detail below), age, time since exposure, and the gender of exposed individuals. The role of these factors in risk estimation is discussed below.

The Effects of Dose

The most fundamental variable in risk estimation is the amount (dose) of radiation to which the individual is exposed. If an epidemiologic study fails to show a relationship between dose and effect for an agent such as radiation, then there is little justification for ascribing the effect to the agent. However, if there is a relationship and the epidemiologic data are sufficiently robust, then a mathematical expression for the relationship between dose and effect can be determined.

The fitting of a mathematical expression to experimental dose–effect data is an interesting undertaking but is of little value unless it leads to other insights. Two goals of this effort are (1) the development of a theory that leads to an understanding of mechanisms and (2) extrapolation of risk into dose regimens that, for one reason or another, cannot be measured. The second goal has attracted considerable attention over a number of years. Because the stochastic effects of radiation all occur naturally in any human population, it is difficult to identify the magnitude of radiation effects at low dose levels (levels at which the effects of radiation are well below the natural rates of occurrence of the effects). Given the statistical fluctuation of the natural incidence of stochastic effects in a population, the additional contribution of radiation exposure at low levels of dose may not be directly observable in an epidemiologic study. It is precisely at these levels that there is the greatest interest in estimating risk, because the low-dose region encompasses diagnostic radiology, occupational, and background radiation exposures. Thus there is a great interest in choosing the proper functional form for a graph of stochastic effects versus radiation dose.

Generic dose–effect curves are shown in Margin Figure 27-3, and actual dose–effect curves are provided in Margin Figure 27-4. The generic curves demonstrate features that have been identified in many epidemiologic studies. The curves contain a region characterized by an increase in biologic effect as dose is increased. But at even higher doses, the effect actually diminishes because of cell killing. For reasons described earlier, there are no data at very low dose levels.

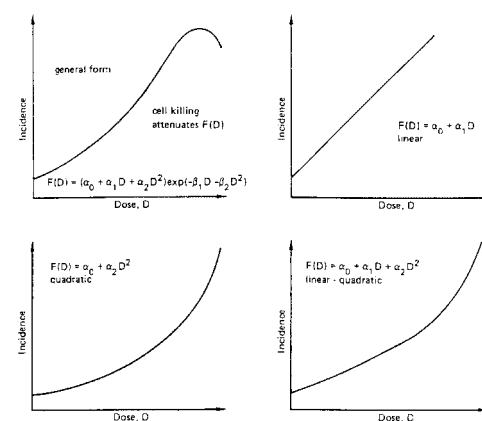
A generally accepted model for the magnitude of radiation effects in a population, $F(D)$, as a function of dose, D , is the equation

$$F(D) = (a_0 + a_1 D + a_2 D^2 + \cdots + a_n D^n) e^{(-b_1 D - b_2 D^2)} \quad (27-1)$$

Polynomial

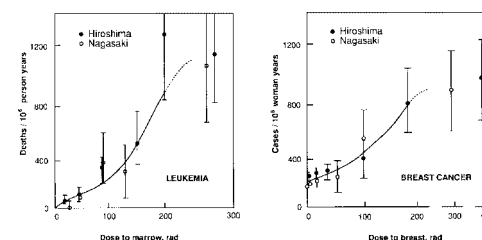
Exponential

Responses to concerns over radioactive fallout in the 1950s and early 1960s led to an effort to reduce exposures from medical radiation. One consequence of this effort was passage of the Radiation Control for Health and Safety Act of 1968. This act established manufacturing performance standards for radiation-emitting devices.



MARGIN FIGURE 27-3

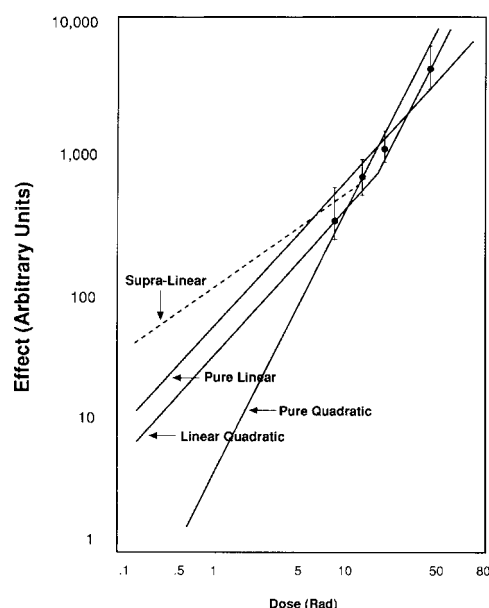
Mathematical forms of dose–effect models for radiation. (From Committee on the Biological Effects on Ionizing Radiation III. *The Effects on Populations of Exposure to Low Levels of Ionizing Radiation*. Washington, D.C., National Academy Press, 1980, p. 28. Used with permission.)



MARGIN FIGURE 27-4

Dose–effect data from atom bomb survivors. (From Straume, T., Dobson, R. L. *Health Phys.* 1981; 41:666–671. Used with permission.)

For the purpose of radiation protection, controlled areas are areas under the direct supervision of a radiation safety officer (RSO), sometimes called a radiation protection officer (RPO). An RSO must be able to document extensive training and experience in radiobiology and radiation safety. The Nuclear Regulatory Commission (NRC) also requires that this individual must have administrative authority of a level that would allow him or her to halt unsafe programs or practices.



MARGIN FIGURE 27-5

Extrapolations of dose–effect curve fits to a low-dose regimen (below the range of reliable data) yield predictions that differ by an order of magnitude or more. (From Edwards, F. M. Dose–Effect Models for Radiation Exposure, in Hendee, W. R. (ed.), *Health Effects of Low-Level Radiation*. East Norwalk, CT, Appleton–Century–Crofts, 1984, p. 87. Used with permission.)

This equation is the product of two functional forms, a polynomial and an exponential. The polynomial is a simple functional form that contains easily interpretable parameters. The coefficients $a_0, a_1, \dots$ are selected, usually by computer search, to provide the best fit to the data. The coefficient a_0 is the constant term, or y-axis intercept, that represents the natural incidence of the disorder (i.e., incidence at zero dose). The coefficient a_1 is the linear coefficient that characterizes the slope of the curve. The “higher-order” coefficients a_2, a_3 , and so on, determine curvature and other detailed elements of the curve.

The exponential portion of Eq. (27-1) has little effect upon the curve if the dose D is small. At high doses (large values of D), the exponential portion illustrates the decrease in stochastic effects due to cell killing. That is, at high doses there are fewer stochastic consequences of radiation exposure because the nonstochastic effects take precedence. Because we are primarily interested in the lower-dose portion of dose–effect curves, examination of the dose–effect curve will be restricted to the polynomial portion.

In general, the polynomial portion of Eq. (27-1) can be shaped to fit the data of a typical dose–effect curve. For a particular data set, a polynomial with the number of coefficients equal to the number of data points can fit the data exactly. However, this is not necessarily desirable. Because there is uncertainty in each data point, the “true” curve would not oscillate to fit each actual point, but would take a straighter path through the points and require only lower-order terms.

There is considerable controversy about which polynomial coefficients should be used to fit epidemiologic data related to the general dose–effect relationship for radiation exposure. The reason for the controversy is that although various coefficients may fit the data at higher doses, the predictions at lower doses, where there are no data (or less reliable data), are different. The controversy centers around whether the polynomial should be linear (coefficients a_0 and a_1 only), linear quadratic (a_0, a_1 , and a_2), or pure quadratic (a_0, a_2 only). Fits to epidemiologic data tend to extrapolate to lower doses in such a way that the pure linear fit predicts the highest effect, pure quadratic predicts the lowest, and the linear-quadratic predicts an intermediate effect. The issue of the best model for dose–effect estimates at low doses is highly significant, because the predictions vary by as much as an order of magnitude at low doses. Currently, there is growing support for using the linear dose–response model for most forms of cancer and for genetic effects, and the linear quadratic model for leukemia.

The Effects of Dose Rate

From basic experiments involving cell cultures and laboratory animals, it is known that dose rate plays a role in determining biologic effects. In the range of total doses over which an effect is observed, the effect often decreases if the dose rate is lowered. Data in human populations support this concept.^{56,57} Indeed, dose rate and dose fractionation are two commonly used principles of radiation therapy. Tissue repair mechanisms generally are not as robust in tumors as in normal tissue. Fractionated radiation therapy treatments are designed to provide normal tissue a chance for repair without significantly reducing the damage to tumor tissue. At low doses where only stochastic processes are expected, there is some uncertainty about the role of dose rate.

The dose rate effectiveness factor (DREF) is defined as

$$\frac{\text{Total dose to elicit an effect at the reference dose rate}}{\text{Total dose to elicit an effect at the dose rate of interest}} = \text{DREF}$$

For example, assume that a given biologic effect is elicited at a particular reference dose rate when the total dose is 1.5 Gy. If the same radiation is delivered at a dose rate with a DREF of 3, then a total dose of only 0.5 Gy would be required to elicit the same effect. Estimates for DREF in humans at low total doses are usually in the range from 2 to 3, although some estimates yield a value as high as 10.⁵⁶